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T cell receptors recognizing MHC class II-restricted MAGE-A3

Abstract

The invention provides an isolated or purified T-cell receptor (TCR) having antigenic specificity for MHC Class II-restricted MAGE-A3. The invention further provides related polypeptides and proteins, as well as related nucleic acids, recombinant expression vectors, host cells, and populations of cells. Further provided by the invention are antibodies, or an antigen binding portion thereof, and pharmaceutical compositions relating to the TCRs of the invention. Methods of detecting the presence of cancer in a host and methods of treating or preventing cancer in a mammal are further provided by the invention.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. application Ser. No. 16/812,845, filed Mar. 9, 2020, which is a divisional of U.S. application Ser. No. 15/848,344, filed Dec. 20, 2017, which issued as U.S. Pat. No. 10,611,815, which is a divisional of U.S. application Ser. No. 14/427,671, which issued as U.S. Pat. No. 9,879,065, which is a U.S. National Phase of International Patent Application No. PCT/US2013/059608, filed Sep. 13, 2013, which claims the benefit of U.S. Provisional Application

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY

(1) Incorporated by reference in its entirety herein is a computer-readable nucleotide/amino acid sequence listing submitted concurrently herewith and identified as follows: One 100,168 Byte XML file named "764792.XML," dated Sep. 28, 2022.

BACKGROUND OF THE INVENTION

(2) Adoptive cell therapy (ACT) involves the transfer of reactive T cells into patients, including the transfer of tumor-reactive T cells into cancer patients. Adoptive cell therapy using T-cells that target human leukocyte antigen (HLA)-A*02 restricted T-cell epitopes has been successful in causing the regression of tumors in some patients. However, patients that lack HLA-A*02 expression cannot be treated with T-cells that target HLA-A*02 restricted T-cell epitopes. Such a limitation creates an obstacle to the widespread application of adoptive cell therapy. Accordingly, there exists a need for improved immunological compositions and methods for treating cancer.

BRIEF SUMMARY OF THE INVENTION

(3) An embodiment of the invention provides an isolated or purified T-cell receptor (TCR), and functional portions and functional variants thereof, having antigenic specificity for MAGE-A3.sub.243-258 and MAGE-A6.

(4) Another embodiment of the invention provides an isolated or purified TCR comprising (a) SEQ ID NOs: 3-8 or (b) SEQ ID NOs: 21-22, or a functional variant of (a) or (b), wherein the functional variant comprises (a) or (b) with at least one amino acid substitution in any one or more of (a) or any one or more of (b), and the functional variant has antigenic specificity for MAGE-A3 in the context of HLA-DPβ1*04.

(5) The invention further provides related polypeptides and proteins, as well as related nucleic acids, recombinant expression vectors, host cells, and populations of cells. Further provided by the invention are antibodies, or antigen binding portions thereof, and pharmaceutical compositions relating to the TCRs (including functional portions and functional variants thereof) of the invention.

(6) Methods of detecting the presence of cancer in a mammal and methods of treating or preventing cancer in a mammal are further provided by the invention. The inventive method of detecting the presence of cancer in a mammal comprises (i) contacting a sample comprising cells of the cancer with any of the inventive TCRs (including functional portions and functional variants thereof), polypeptides, proteins, nucleic acids, recombinant expression vectors, host cells, populations of host cells, or antibodies, or antigen binding portions thereof, described herein, thereby forming a complex, and (ii) detecting the complex, wherein detection of the complex is indicative of the presence of cancer in the mammal.

(7) The inventive method of treating or preventing cancer in a mammal comprises administering to the mammal any of the TCRs (including functional portions and functional variants thereof), polypeptides, or proteins described herein, any nucleic acid or recombinant expression vector comprising a nucleotide sequence encoding any of the TCRs (including functional portions and functional variants thereof), polypeptides, proteins described herein, or any host cell or population of host cells comprising a recombinant vector which encodes any of the TCRs (including functional portions and functional variants thereof), polypeptides, or proteins described herein, in an amount effective to treat or prevent cancer in the mammal.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) FIGS. 1A and 1B are bar graphs showing interferon (IFN)-gamma secretion (pg/ml) by CD4+ T

cells from first (1A) and second (1B) donors in response to co-culture with 293-CIITA target cells untransfected (292-CIITA) or transfected with full length NY-ESO-1 (293-CIITA-NY-ESO-1) protein, MAGE-A1 protein (293-CIITA-MAGE A1), MAGE-A3 protein (293-CIITA-MAGE-A3), MAGE-A6 protein (293-CIITA-MAGE-A6), or MAGE A12 protein (293-CIITA-MAGE-A12). The T cells were untransduced (UT) or transduced with F5 (anti-MART-1) TCR, R12C9 TCR, or 6F9 TCR.

(2) FIG. 2A is a bar graph showing IFN-gamma secretion (pg/ml) by T cells from a human donor that were untransduced or transduced with 6F9 TCR or F5 TCR in response to co-culture with 526-CIITA cells that were untreated or treated with anti-MAGE-A3 siRNA or anti-MART-1 siRNA.

(3) FIG. 2B is a bar graph showing IFN-gamma secretion (pg/ml) by CD4⁺ T cells from a human donor that were untransduced or transduced with 6F9 TCR in response to co-culture with H1299-CIITA cells that were untreated or treated with anti-MAGE-A3 siRNA or anti-MART-1 siRNA.

(4) FIG. 3 is a bar graph showing IFN-gamma secretion (pg/ml) by 6F9-transduced PBL cultured alone (T cells only) or co-cultured with 3071 cells, 3071-CIITA cells, 397 cells, 397-CIITA cells, 2630 cells, 2630-CIITA cells, 2984 cells, or 2984-CIITA cells.

(5) FIG. 4 is a bar graph showing IFN-gamma secretion (pg/ml) by CD4⁺ enriched PBL that were transduced with 6F9 TCR or untransduced upon culture alone (T cell only) or in response to co-culture with untreated H1299-CIITA cells, H1299-CIITA transfected with anti-HLA-DP or anti-HLA-DR siRNA, untreated 526-CIITA cells, or 526-CIITA transfected with anti-HLA-DP or anti-HLA-DR siRNA.

(6) FIG. 5 is a bar graph showing IFN-gamma (pg/ml) secretion by PBL that were untransduced or transduced with wild-type (wt) 6F9 TCR or one of each of eight substituted TCRs: a1 (alpha chain S116A), a2 (alpha chain S117A), a3 (alpha chain G118A), a4 (alpha chain T119A), b1 (beta chain R115A), b2 (beta chain T116A), b3 (beta chain G117A), or b4 (beta chain P118A) upon culture alone (T cell only; unshaded bars) or co-culture with 624-CIITA (checkered bars), 526-CIITA (right crosshatched bars), 1359-CIITA (horizontal striped bars), H1299-CIITA (left crosshatched bars), or 1764-CIITA (vertical striped bars).

(7) FIG. 6 is a bar graph showing IFN-gamma (pg/ml) secretion by CD4⁺ enriched PBL that were untransduced or transduced with wild-type (wt) 6F9 TCR or one of each of three substituted TCRs: a1 (alpha chain S116A), a2 (alpha chain S117A), or b2 (beta chain T116A) upon culture alone (T cell only; unshaded bars) or co-culture with 624-CIITA (checkered bars), 526-CIITA (right crosshatched bars), 1359-CIITA (horizontal striped bars), H1299-CIITA (left crosshatched bars), or 1764-CIITA (vertical striped bars).

(8) FIG. 7 is a bar graph showing IFN-gamma (pg/ml) secretion by PBL that were untransduced or transduced with wild-type (wt) 6F9 TCR or one of each of ten substituted TCRs: a1 (alpha chain S116A), a2 (alpha chain S117A), a1-1 (alpha chain S116L), a1-2 (alpha chain S116I), a1-3 (alpha chain S116V), a1-4 (alpha chain S116M), a2-1 (alpha chain S117L), a2-2 (alpha chain S117I), a2-3 (alpha chain S117V), or a2-4 (alpha chain S117M) upon culture alone or (T cell only; unshaded bars) or co-culture with 624-CIITA (right crosshatched bars), 526-CIITA (vertical striped bars), 1359-CIITA (horizontal striped bars), H1299-CIITA (left crosshatched bars), or 1764-CIITA (black bars).

(9) FIG. 8 is a bar graph showing IFN-gamma (pg/ml) secretion by PBL that were untransduced (checkered bars) or transduced with wild-type (wt) 6F9 TCR (horizontal striped bar) or 6F9mC TCR (SEQ ID NOs: 27 and 28) (left crosshatched bars) upon culture alone (T cells only) or co-culture with 624-CIITA, 1300-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, 397-CIITA, 2630-CIITA, 2984-CIITA, 3071-CIITA, or 1764-CIITA cells.

(10) FIGS. 9A and 9B are bar graphs showing IFN-gamma (pg/ml) secretion by CD4⁺ (9A) or CD8⁺ (9B) enriched PBL that were untransduced (checkered bars) or transduced with wild-type (wt) 6F9 TCR (horizontal striped bar) or 6F9mC TCR (SEQ ID NOs: 27 and 28) (left crosshatched bars) upon culture alone (T cells only) or co-culture with 624-CIITA, SK37-CIITA, 526-CIITA,

1359-CIITA, H1299-CIITA, 397-CIITA, 2630-CIITA, 2984-CIITA, 3071-CIITA, or 1764-CIITA cells.

(11) FIG. 10A is a bar graph showing IFN-gamma secretion by PBL that were untransduced (UT; unshaded bars) or transduced with R12C9 TCR (grey bars) or 6F9 TCR (black bars) upon culture alone (none) or co-culture with 293-CIITA transfectants that were transfected with full length NY-ESO-1 protein, MAGE-A1 protein, MAGE-A3 protein, MAGE-A6 protein, MAGE-A12 protein, or 293-CIITA cells that were pulsed with MAGE-A 3.sub.243-258 peptide or MAGE-A3 protein.

(12) FIG. 10B is a bar graph showing IFN-gamma secretion by PBL that were untransduced (UT; black bars) or transduced with 6F9 TCR (grey bars) upon culture alone (T cell alone) or co-culture with non-small cell lung cancer (NSCLC) cell line H11299 or melanoma cell line 526 mel, 624 mel, or 1359 mel.

DETAILED DESCRIPTION OF THE INVENTION

(13) The invention provides an isolated or purified T cell receptor (TCR), and functional portions and functional variants thereof, having antigenic specificity for MAGE-A3, wherein the TCR recognizes MAGE-A3 in the context of HLA-DP β 1*04. In an embodiment of the invention, the isolated or purified TCR has antigenic specificity for MAGE-A3.sub.243-258 and MAGE-A6.

(14) MAGE-A3 and MAGE-A6 are members of the MAGE-A family of twelve homologous proteins, also including MAGE-A1, MAGE-A2, MAGE-A4, MAGE-A5, MAGE-A7, MAGE-A8, MAGE-A9, MAGE-A10, MAGE-A11, and MAGE-A12. The MAGE-A proteins are cancer testis antigens (CTA), which are expressed only in tumor cells and non-MHC expressing germ cells of the testis and placenta. MAGE-A proteins are expressed in a variety of human cancers including, but not limited to, melanoma, breast cancer, leukemia, thyroid cancer, gastric cancer, pancreatic cancer, liver cancer (e.g., hepatocellular carcinoma), lung cancer (e.g., non-small cell lung carcinoma), ovarian cancer, multiple myeloma, esophageal cancer, kidney cancer, head cancers (e.g., squamous cell carcinoma), neck cancers (e.g., squamous cell carcinoma), prostate cancer, synovial cell sarcoma, and urothelial cancer.

(15) The TCRs (including functional portions and functional variants thereof) of the invention provide many advantages, including when used for adoptive cell transfer. For example, by targeting MAGE-A3 that is presented in the context of HLA-DP β 1*04, the inventive TCRs (including functional portions and functional variants thereof) make it possible to treat patients who are unable to be treated using TCRs that target MAGE antigens that are presented in the context of other HLA molecules, e.g., HLA-A*02, HLA-A*01, or HLA-C*07. HLA-DP β 1*04 is a highly prevalent allele that is expressed by from about 70% to about 80% of the cancer patient population. Accordingly, the inventive TCRs (including functional portions and functional variants thereof) advantageously greatly expand the patient population that can be treated. Additionally, without being bound by a particular theory, it is believed that because MAGE-A3 and/or MAGE-A6 are expressed by cells of multiple cancer types, the inventive TCRs (including functional portions and functional variants thereof) advantageously provide the ability to destroy cells of multiple types of cancer and, accordingly, treat or prevent multiple types of cancer. Additionally, without being bound to a particular theory, it is believed that because the MAGE-A proteins are cancer testis antigens that are expressed only in tumor cells and non-MHC expressing germ cells of the testis and placenta, the inventive TCRs (including functional portions and functional variants thereof) advantageously target the destruction of cancer cells while minimizing or eliminating the destruction of normal, non-cancerous cells, thereby reducing, for example, by minimizing or eliminating, toxicity.

(16) The phrase “antigenic specificity” as used herein means that the TCR can specifically bind to and immunologically recognize MAGE-A3 and/or MAGE-A6 with high avidity. For example, a TCR may be considered to have “antigenic specificity” for MAGE-A3 and/or MAGE-A6 if T cells expressing the TCR secrete at least about 200 pg/ml or more (e.g., 200 pg/ml or more, 300 pg/ml or more, 400 pg/ml or more, 500 pg/ml or more, 600 pg/ml or more, 700 pg/ml or more, 1000 pg/ml or more, 5,000 pg/ml or more, 7,000 pg/ml or more, 10,000 pg/ml or more, or 20,000 pg/ml or

more) of IFN- γ upon co-culture with antigen-negative HLA-DP β 1*04+ target cells pulsed with a low concentration of MAGE-A3 and/or MAGE-A6 peptide (e.g., about 0.05 ng/ml to about 5 ng/ml, 0.05 ng/ml, 0.1 ng/ml, 0.5 ng/ml, 1 ng/ml, or 5 ng/ml). Alternatively or additionally, a TCR may be considered to have “antigenic specificity” for MAGE-A3 and/or MAGE-A6 if T cells expressing the TCR secrete at least twice as much IFN- γ as the untransduced PBL background level of IFN- γ upon co-culture with antigen-negative HLA-DP β 1*04+ target cells pulsed with a low concentration of MAGE-A3 and/or MAGE-A6 peptide. The inventive TCRs (including functional portions and functional variants thereof) may also secrete IFN- γ upon co-culture with antigen-negative HLA-DP β 1*04+ target cells pulsed with higher concentrations of MAGE-A3 and/or MAGE-A6 peptide.

(17) An embodiment of the invention provides a TCR (including functional portions and variants thereof) with antigenic specificity for any MAGE-A3 protein, polypeptide or peptide. The inventive TCR (including functional portions and functional variants thereof) may have antigenic specificity for a MAGE-A3 protein comprising, consisting of, or consisting essentially of, SEQ ID NO: 1. In a preferred embodiment of the invention, the TCR (including functional portions and variants thereof) has antigenic specificity for a MAGE-A3.sub.243-258 peptide comprising, consisting of, or consisting essentially of, KKLLTQHFVQENYLEY (SEQ ID NO: 2).

(18) The inventive TCRs (including functional portions and functional variants thereof) are able to recognize MAGE-A3 in a human leukocyte antigen (HLA)-DP β 1*04-dependent manner. “HLA-DP β 1*04-dependent manner,” as used herein, means that the TCR elicits an immune response upon binding to a MAGE-A3 protein, polypeptide or peptide within the context of an HLA-DP β 1*04 molecule. The inventive TCRs (including functional portions and functional variants thereof) are able to recognize MAGE-A3 that is presented by an HLA-DP β 1*04 molecule and may bind to the HLA-DP β 1*04 molecule in addition to MAGE-A3. Exemplary HLA-DP β 1*04 molecules, in the context of which the inventive TCRs (including functional portions and functional variants thereof) recognize MAGE-A3, include those encoded by the HLA-DP β 1*0401 and/or HLA-DP β 1*0402 alleles.

(19) An embodiment of the invention provides a TCR (including functional portions and variants thereof) with antigenic specificity for any MAGE-A6 protein, polypeptide or peptide. The inventive TCR (including functional portions and functional variants thereof) may have antigenic specificity for a MAGE-A6 protein comprising, consisting of, or consisting essentially of, SEQ ID NO: 45. In a preferred embodiment of the invention, the TCR (including functional portions and functional variants thereof) has antigenic specificity for a MAGE-A6.sub.243-258 peptide comprising, consisting of, or consisting essentially of, KKLLTQYFVQENYLEY (SEQ ID NO: 46).

(20) The invention provides a TCR comprising two polypeptides (i.e., polypeptide chains), such as an alpha (α) chain of a TCR, a beta (β) chain of a TCR, a gamma (γ) chain of a TCR, a delta (δ) chain of a TCR, or a combination thereof. The polypeptides of the inventive TCR can comprise any amino acid sequence, provided that the TCR has antigenic specificity for MAGE-A3 in the context of HLA-DP β 1*04.

(21) In an embodiment of the invention, the TCR comprises two polypeptide chains, each of which comprises a variable region comprising a complementarity determining region (CDR)1, a CDR2, and a CDR3 of a TCR. In an embodiment of the invention, the TCR comprises a first polypeptide chain comprising a CDR1 comprising the amino acid sequence of SEQ ID NO: 3 or 13 (CDR1 of a chain), a CDR2 comprising the amino acid sequence of SEQ ID NO: 4 or 14 (CDR2 of a chain), and a CDR3 comprising the amino acid sequence of SEQ ID NO: 5 or 15 (CDR3 of a chain), and a second polypeptide chain comprising a CDR1 comprising the amino acid sequence of SEQ ID NO: 6 or 16 (CDR1 of β chain), a CDR2 comprising the amino acid sequence of SEQ ID NO: 7 or 17 (CDR2 of β chain), and a CDR3 comprising the amino acid sequence of SEQ ID NO: 8 or 18 (CDR3 of β chain). In this regard, the inventive TCR can comprise any one or more of the amino

acid sequences selected from the group consisting of any one or more of SEQ ID NOs: 3-5, 6-8, 13-15, and 16-18. Preferably the TCR comprises the amino acid sequences of SEQ ID NOs: 3-8 or 13-18. More preferably the TCR comprises the amino acid sequences of SEQ ID NOs: 3-8.

(22) Alternatively or additionally, the TCR can comprise an amino acid sequence of a variable region of a TCR comprising the CDRs set forth above. In this regard, the TCR can comprise the amino acid sequence of SEQ ID NO: 9 or 19 (the variable region of an α chain) or 10 or 20 (the variable region of a β chain), both SEQ ID NOs: 9 and 10 or both SEQ ID NOs: 19 and 20.

Preferably, the inventive TCR comprises the amino acid sequences of both SEQ ID NOs: 9 and 10.

(23) Alternatively or additionally, the TCR can comprise an α chain of a TCR and a β chain of a TCR. Each of the α chain and β chain of the inventive TCR can independently comprise any amino acid sequence. Preferably, the α chain comprises the variable region of an α chain as set forth above. In this regard, the inventive TCR can comprise the amino acid sequence of SEQ ID NO: 11 or 21. An α chain of this type can be paired with any β chain of a TCR. Preferably, the β chain of the inventive TCR comprises the variable region of a β chain as set forth above. In this regard, the inventive TCR can comprise the amino acid sequence of SEQ ID NO: 12 or 22. The inventive TCR, therefore, can comprise the amino acid sequence of SEQ ID NO: 11, 12, 21, or 22, both SEQ ID NOs: 11 and 12 or both SEQ ID NOs: 21 and 22. Preferably, the inventive TCR comprises the amino acid sequences of both SEQ ID NOs: 11 and 12.

(24) Included in the scope of the invention are functional variants of the inventive TCRs described herein. The term "functional variant" as used herein refers to a TCR, polypeptide, or protein having substantial or significant sequence identity or similarity to a parent TCR, polypeptide, or protein, which functional variant retains the biological activity of the TCR, polypeptide, or protein of which it is a variant. Functional variants encompass, for example, those variants of the TCR, polypeptide, or protein described herein (the parent TCR, polypeptide, or protein) that retain the ability to specifically bind to MAGE-A3 and/or MAGE-A6 for which the parent TCR has antigenic specificity or to which the parent polypeptide or protein specifically binds, to a similar extent, the same extent, or to a higher extent, as the parent TCR, polypeptide, or protein. In reference to the parent TCR, polypeptide, or protein, the functional variant can, for instance, be at least about 30%, 50%, 75%, 80%, 90%, 95%, 96%, 97%, 98%, 99% or more identical in amino acid sequence to the parent TCR, polypeptide, or protein.

(25) The functional variant can, for example, comprise the amino acid sequence of the parent TCR, polypeptide, or protein with at least one conservative amino acid substitution. Conservative amino acid substitutions are known in the art, and include amino acid substitutions in which one amino acid having certain physical and/or chemical properties is exchanged for another amino acid that has the same chemical or physical properties. For instance, the conservative amino acid substitution can be an acidic amino acid substituted for another acidic amino acid (e.g., Asp or Glu), an amino acid with a nonpolar side chain substituted for another amino acid with a nonpolar side chain (e.g., Ala, Gly, Val, Ile, Leu, Met, Phe, Pro, Trp, Val, etc.), a basic amino acid substituted for another basic amino acid (Lys, Arg, etc.), an amino acid with a polar side chain substituted for another amino acid with a polar side chain (Asn, Cys, Gln, Ser, Thr, Tyr, etc.), etc.

(26) Alternatively or additionally, the functional variants can comprise the amino acid sequence of the parent TCR, polypeptide, or protein with at least one non-conservative amino acid substitution. In this case, it is preferable for the non-conservative amino acid substitution to not interfere with or inhibit the biological activity of the functional variant. Preferably, the non-conservative amino acid substitution enhances the biological activity of the functional variant, such that the biological activity of the functional variant is increased as compared to the parent TCR, polypeptide, or protein.

(27) In this regard, an embodiment of the invention provides an isolated or purified TCR comprising (a) SEQ ID NOs: 3-8, (b) SEQ ID NOs: 21-22, or a functional variant of (a) or (b), wherein the functional variant comprises (a) or (b) with at least one amino acid substitution in any

one or more of (a) or any one or more of (b), and the functional variant has antigenic specificity for MAGE-A3 in the context of HLA-DPβ1*04. Preferably, the amino acid substitution is located in a CDR3 region of the alpha or beta chain, preferably in the CDR3 region of the alpha chain. In some embodiments, the functional variant (or functional portions thereof) provide an increased reactivity against MAGE-A3 as compared to the parent TCR amino acid sequence. In general, the substituted α chain amino acid sequences SEQ ID NOs: 29, 31, and 33 correspond with all or portions of the native, unsubstituted SEQ ID NO: 11 (TCR α chain), with SEQ ID NOs: 29, 31, and 33 having at least one substitution when compared to SEQ ID NO: 11. Preferably, one or more of the native Ser116, Ser117, Gly118, and Thr119 is substituted. Likewise, the substituted 13 chain amino acid sequences SEQ ID NOs: 30, 32, and 34 correspond with all or portions of the native, unsubstituted SEQ ID NO: 12 (TCR β chain), with SEQ ID NOs: 30, 32, and 34 having at least one substitution when compared to SEQ ID NO: 12. Preferably, one or more of the native Arg115, Thr116, Gly117, and Pro118 is substituted.

(28) In particular, the invention provides a functional variant of a TCR comprising (i) SEQ ID NO: 29, wherein Xaa4 is Ser, Ala, Leu, Ile, Val, or Met; Xaa5 is Ser, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Thr, Ala, Leu, Ile, Val, or Met and/or (ii) SEQ ID NO: 30, wherein Xaa4 is Arg, Ala, Leu, Ile, Val, or Met; Xaa5 is Thr, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Pro, Ala, Leu, Ile, Val, or Met. SEQ ID NO: 29 generally corresponds to positions 113-123 of the native, unsubstituted SEQ ID NO: 11 with the exception that in SEQ ID NO: 29, one or more of Ser4, Ser5, Gly6, and Thr7 is substituted. Preferably, the functional variant comprises (a) SEQ ID NO: 29, wherein Xaa4 is Ala, Xaa5 is Ser, Xaa6 is Gly, and Xaa7 is Thr, or (b) SEQ ID NO: 29, wherein Xaa4 is Ser, Xaa5 is Ala, Xaa6 is Gly, and Xaa7 is Thr. Although in some embodiments, SEQ ID NO: 29 may comprise wild-type CDR3α SEQ ID NO: 5, preferably, SEQ ID NO: 29 does not comprise SEQ ID NO: 5. SEQ ID NO: 30 generally corresponds to positions 112-126 of the native, unsubstituted SEQ ID NO: 12 with the exception that in SEQ ID NO: 30, one or more of Arg4, Thr5, Gly6, and Pro7 is substituted. Although in some embodiments, SEQ ID NO: 30 may comprise wild-type CDR3β SEQ ID NO: 8, preferably, SEQ ID NO: 30 does not comprise SEQ ID NO: 8.

(29) The invention also provides a functional variant of a TCR comprising (i) SEQ ID NO: 31, wherein Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; and/or (ii) SEQ ID NO: 32, wherein Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met. SEQ ID NO: 31 generally corresponds to positions 1-134 of the native, unsubstituted SEQ ID NO: 11 with the exception that in SEQ ID NO: 31, one or more of one or more of Ser116, Ser117, Gly118, and Thr119 is substituted. Preferably, the functional variant comprises (a) SEQ ID NO: 31, wherein Xaa116 is Ala, Xaa117 is Ser, Xaa118 is Gly, and Xaa119 is Thr, or (b) SEQ ID NO: 31, wherein Xaa116 is Ser, Xaa117 is Ala, Xaa118 is Gly, and Xaa119 is Thr. Although in some embodiments, SEQ ID NO: 31 may comprise wild-type CDR3α SEQ ID NO: 5, preferably, SEQ ID NO: 31 does not comprise SEQ ID NO: 5. SEQ ID NO: 32 generally corresponds to positions 1-137 of the native, unsubstituted SEQ ID NO: 12 with the exception that in SEQ ID NO: 32, one or more of one or more of Arg115, Thr116, Gly117, and Pro118 is substituted. Although in some embodiments, SEQ ID NO: 32 may comprise wild-type CDR3β SEQ ID NO: 8, preferably, SEQ ID NO: 32 does not comprise SEQ ID NO: 8.

(30) Also provided by the invention is functional variant of a TCR comprising (i) SEQ ID NO: 33, wherein Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; and/or (ii) SEQ ID NO: 34, wherein Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met. SEQ ID NO: 33 generally corresponds to positions 1-275 of the native, unsubstituted SEQ ID NO: 11

with the exception that in SEQ ID NO: 33, one or more of one or more of Ser116, Ser117, Gly118, and Thr119 is substituted. Preferably, the functional variant comprises (a) SEQ ID NO: 33, wherein Xaa116 is Ala, Xaa117 is Ser, Xaa118 is Gly, and Xaa119 is Thr, or (b) SEQ ID NO: 33, wherein Xaa116 is Ser, Xaa117 is Ala, Xaa118 is Gly, and Xaa119 is Thr. Although in some embodiments, SEQ ID NO: 33 may comprise wild-type CDR3 α SEQ ID NO: 5, preferably, SEQ ID NO: 33 does not comprise SEQ ID NO: 5. SEQ ID NO: 34 generally corresponds to positions 1-313 of the native, unsubstituted SEQ ID NO: 12 with the exception that in SEQ ID NO: 34, one or more of one or more of Arg115, Thr116, Gly117, and Pro118 is substituted. Although in some embodiments, SEQ ID NO: 34 may comprise wild-type CDR3 β SEQ ID NO: 8, preferably, SEQ ID NO: 34 does not comprise SEQ ID NO: 8.

(31) Like the TCRs of the invention, the functional variants described herein comprise two polypeptide chains, each of which comprises a variable region comprising a CDR1, a CDR2, and a CDR3 of a TCR. Preferably, the first polypeptide chain comprises a CDR1 comprising the amino acid sequence of SEQ ID NO: 3 (CDR1 of α chain), a CDR2 comprising the amino acid sequence of SEQ ID NO: 4 (CDR2 of α chain), and a substituted CDR3 comprising the amino acid sequence of SEQ ID NO: 29 (substituted CDR3 of α chain), and the second polypeptide chain comprises a CDR1 comprising the amino acid sequence of SEQ ID NO: 6 (CDR1 of β chain), a CDR2 comprising the amino acid sequence of SEQ ID NO: 7 (CDR2 of β chain), and a CDR3 comprising the amino acid sequence of SEQ ID NO: 8 (CDR3 of β chain). In another embodiment, the first polypeptide chain comprises a CDR1 comprising the amino acid sequence of SEQ ID NO: 3 (CDR1 of a chain), a CDR2 comprising the amino acid sequence of SEQ ID NO: 4 (CDR2 of a chain), and a CDR3 comprising the amino acid sequence of SEQ ID NO: 5 (CDR3 of α chain), and the second polypeptide chain comprises a CDR1 comprising the amino acid sequence of SEQ ID NO: 6 (CDR1 of β chain), a CDR2 comprising the amino acid sequence of SEQ ID NO: 7 (CDR2 of β chain), and a substituted CDR3 comprising the amino acid sequence of SEQ ID NO: 30 (substituted CDR3 of β chain). In this regard, the inventive functional variant of a TCR can comprise the amino acid sequences selected from the group consisting of SEQ ID NOs: 3-5; SEQ ID NOs: 3-4 and 29; SEQ ID NOs: 6-8; and SEQ ID NOs: 6-7 and 30. Preferably the functional variant of a TCR comprises the amino acid sequences of SEQ ID NOs: 3-4, 29, and 6-8; SEQ ID NOs: 3-7 and 30; or SEQ ID NOs: 3-4, 29, 6-7, and 30. More preferably, the functional variant of a TCR comprises the amino acid sequences of SEQ ID NOs: 3-4, 29, and 6-8.

(32) Alternatively or additionally, the functional variant of a TCR can comprise a substituted amino acid sequence of a variable region of a TCR comprising the CDRs set forth above. In this regard, the TCR can comprise the substituted amino acid sequence of SEQ ID NO: 31 (the substituted variable region of an α chain), 10 (the variable region of a β chain), both SEQ ID NOs: 31 and 10, the substituted amino acid sequence of SEQ ID NO: 32 (the substituted variable region of an β chain), 9 (the variable region of an α chain), both SEQ ID NOs: 9 and 32, or both SEQ ID NOs: 31 and 32. Preferably, the inventive functional variant of a TCR comprises the amino acid sequences of SEQ ID NOs: 31 and 10 or SEQ ID NOs: 32 and 9. More preferably, the inventive functional variant of a TCR comprises the amino acid sequences of SEQ ID NOs: 31 and 10.

(33) Alternatively or additionally, the functional variant of a TCR can comprise a substituted α chain of a TCR and a β chain of a TCR. Each of the α chain and β chain of the inventive TCR can independently comprise any amino acid sequence. Preferably, the substituted α chain comprises a substituted variable region of an α chain as set forth above. In this regard, the inventive substituted α chain of the TCR can comprise the amino acid sequence of SEQ ID NO: 33. An inventive substituted α chain of this type can be paired with any β chain of a TCR. Preferably, the β chain of the inventive TCR comprises the variable region of a β chain as set forth above. In this regard, the inventive TCR can comprise the amino acid sequence of SEQ ID NO: 12 or the substituted amino acid sequence SEQ ID NO: 34. An inventive substituted β chain of this type can be paired with any α chain of a TCR. In this regard, the inventive TCR can comprise the amino acid sequence of SEQ

ID NO: 11 or 33. The inventive functional variant of a TCR, therefore, can comprise the amino acid sequence of SEQ ID NO: 11, 12, 33, 34, both SEQ ID NOs: 33 and 34; both SEQ ID NOs: 11 and 34; or both SEQ ID NOs: 12 and 33. Preferably, the inventive functional variant of a TCR comprises the amino acid sequences of SEQ ID NOs: 11 and 34 or SEQ ID NOs: 12 and 33. More preferably, the functional variant of a TCR comprises the amino acid sequences of SEQ ID NOs: 12 and 33.

(34) In an embodiment of the invention, the TCR (or functional variant thereof) may comprise a human constant region. In this regard, the TCR (or functional variant thereof) can comprise a human constant region comprising SEQ ID NO: 23 or 35 (human constant region of an α chain), SEQ ID NO: 24 or 36 (human constant region of β chain), both SEQ ID NOs: 23 and 24, or both SEQ ID NOs: 35 and 36. Preferably, the TCR (or functional variant thereof) comprises both SEQ ID NOs: 23 and 24.

(35) In another embodiment of the invention, the TCR (or functional variant thereof) can comprise a human/mouse chimeric TCR (or functional variant thereof). In this regard, the TCR (or functional variant thereof) can comprise a mouse constant region comprising SEQ ID NO: 25 (mouse constant region of an α chain), SEQ ID NO: 26 (mouse constant region of β chain), or both SEQ ID NOs: 25 and 26. Preferably, the TCR (or functional variant thereof) comprises both SEQ ID NOs: 25 and 26.

(36) The inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise any of the CDRs set forth above. In this regard, the inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise the amino acid sequences selected from the group consisting of SEQ ID NOs: 3-5; SEQ ID NOs: 13-15; SEQ ID NOs: 16-18; SEQ ID NOs: 3-4 and 29; SEQ ID NOs: 6-8; and SEQ ID NOs: 6-7 and 30. Preferably the human/mouse chimeric TCR (or functional variant or functional portion thereof) comprises the amino acid sequences of SEQ ID NOs: 3-8; SEQ ID NOs: 13-18; SEQ ID NOs: 3-4, 29, and 6-8; SEQ ID NOs: 3-7 and 30; or SEQ ID NOs: 3-4, 29, 6-7, and 30. More preferably, the human/mouse chimeric TCR (or functional variant or functional portion thereof) comprises the amino acid sequences of SEQ ID NOs: 3-4, 29, and 6-8 or SEQ ID NOs: 3-8.

(37) Alternatively or additionally, the human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise any of the variable regions set forth above. In this regard, the inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise the substituted amino acid sequence of SEQ ID NO: 31 (the substituted variable region of an α chain), SEQ ID NO: 10 or 20 (the variable region of a β chain), the substituted amino acid sequence of SEQ ID NO: 32 (the substituted variable region of an β chain), SEQ ID NO: 9 or 19 (the variable region of an α chain), both SEQ ID NOs: 9 and 32, both SEQ ID NOs: 31 and 32, both SEQ ID NOs: 31 and 10, both SEQ ID NOs: 9 and 10, or both SEQ ID NOs: 19 and 20. Preferably, the inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) comprises the amino acid sequences of SEQ ID NOs: 31 and 10, SEQ ID NOs: 9 and 10, or SEQ ID NOs: 32 and 9. More preferably, the inventive functional variant or functional portion of a TCR comprises the amino acid sequences of SEQ ID NOs: 31 and 10 or SEQ ID NOs: 9 and 10.

(38) Alternatively or additionally, the human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise an α chain of a TCR (or functional variant or functional portion thereof) and a β chain of a TCR (or functional variant or functional portion thereof). Each of the α chain and β chain of the inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) can independently comprise any amino acid sequence. Preferably, the α chain comprises the variable region of an α chain as set forth above. In this regard, the inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise the amino acid sequence of SEQ ID NO: 27. An inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) of this type can be paired with any β chain of a TCR (or functional variant or functional portion thereof). Preferably, the β chain of the inventive

human/mouse chimeric TCR (or functional variant or functional portion thereof) comprises the variable region of a β chain as set forth above. In this regard, the inventive human/mouse chimeric TCR (or functional variant or functional portion thereof) can comprise the amino acid sequence of SEQ ID NO: 28. The inventive human/mouse chimeric TCR (or functional variant or functional portion thereof), therefore, can comprise the amino acid sequence of SEQ ID NO: 27 or 28, or both SEQ ID NOs: 27 and 28. Preferably, the inventive TCR comprises the amino acid sequences of SEQ ID NOs: 27 and 28.

(39) Also provided by the invention is a polypeptide comprising a functional portion of any of the TCRs or functional variants described herein. The term “polypeptide” as used herein includes oligopeptides and refers to a single chain of amino acids connected by one or more peptide bonds.

(40) With respect to the inventive polypeptides, the functional portion can be any portion comprising contiguous amino acids of the TCR (or functional variant thereof) of which it is a part, provided that the functional portion specifically binds to MAGE-A3 and/or MAGE-A6. The term “functional portion” when used in reference to a TCR (or functional variant thereof) refers to any part or fragment of the TCR (or functional variant thereof) of the invention, which part or fragment retains the biological activity of the TCR (or functional variant thereof) of which it is a part (the parent TCR or parent functional variant thereof). Functional portions encompass, for example, those parts of a TCR (or functional variant thereof) that retain the ability to specifically bind to MAGE-A3 (e.g., in an HLA-DP β 1*04-dependent manner) or MAGE-A6, or detect, treat, or prevent cancer, to a similar extent, the same extent, or to a higher extent, as the parent TCR (or functional variant thereof). In reference to the parent TCR (or functional variant thereof), the functional portion can comprise, for instance, about 10%, 25%, 30%, 50%, 68%, 80%, 90%, 95%, or more, of the parent TCR (or functional variant thereof).

(41) The functional portion can comprise additional amino acids at the amino or carboxy terminus of the portion, or at both termini, which additional amino acids are not found in the amino acid sequence of the parent TCR or functional variant thereof. Desirably, the additional amino acids do not interfere with the biological function of the functional portion, e.g., specifically binding to MAGE-A3 and/or MAGE-A6; and/or having the ability to detect cancer, treat or prevent cancer, etc. More desirably, the additional amino acids enhance the biological activity, as compared to the biological activity of the parent TCR or functional variant thereof.

(42) The polypeptide can comprise a functional portion of either or both of the α and β chains of the TCRs or functional variant thereof of the invention, such as a functional portion comprising one or more of CDR1, CDR2, and CDR3 of the variable region(s) of the α chain and/or β chain of a TCR or functional variant thereof of the invention. In this regard, the polypeptide can comprise a functional portion comprising the amino acid sequence of SEQ ID NO: 3 or 13 (CDR1 of α chain), 4 or 14 (CDR2 of α chain), 5, 15, or 29 (CDR3 of α chain), 6 or 16 (CDR1 of β chain), 7 or 17 (CDR2 of β chain), 8, 18, or 30 (CDR3 of β chain), or a combination thereof. Preferably, the inventive polypeptide comprises a functional portion comprising SEQ ID NOs: 3-5; 3-4 and 29; 6-8; 6-7 and 30; 13-15; 16-18; all of SEQ ID NOs: 3-8; all of SEQ ID NOs: 13-18; all of SEQ ID NOs: 3-4, 29, and 6-8; all of SEQ ID NOs: 3-7 and 30; or all of SEQ ID NOs: 3-4, 29, 6-7, and 30. More preferably, the polypeptide comprises a functional portion comprising the amino acid sequences of all of SEQ ID NOs: 3-8 or all of SEQ ID NOs: 3-4, 29, and 6-8.

(43) Alternatively or additionally, the inventive polypeptide can comprise, for instance, the variable region of the inventive TCR or functional variant thereof comprising a combination of the CDR regions set forth above. In this regard, the polypeptide can comprise the amino acid sequence of SEQ ID NO: 9, 19, or 31 (the variable region of an α chain), SEQ ID NO: 10, 20, or 32 (the variable region of a β chain), both SEQ ID NOs: 9 and 10, both SEQ ID NOs: 19 and 20; both SEQ ID NOs: 31 and 32; both SEQ ID NOs: 9 and 32; or both SEQ ID NOs: 10 and 31. Preferably, the polypeptide comprises the amino acid sequences of both SEQ ID NOs: 9 and 10 or both SEQ ID NOs: 10 and 31.

(44) Alternatively or additionally, the inventive polypeptide can comprise the entire length of an α or β chain of one of the TCRs or functional variant thereof described herein. In this regard, the inventive polypeptide can comprise an amino acid sequence of SEQ ID NOs: 11, 12, 21, 22, 27, 28, 33, or 34. Alternatively, the polypeptide of the invention can comprise α and β chains of the TCRs or functional variants thereof described herein. For example, the inventive polypeptide can comprise the amino acid sequences of both SEQ ID NOs: 11 and 12, both SEQ ID NOs: 21 and 22, both SEQ ID NOs: 33 and 34, both SEQ ID NOs: 11 and 34, both SEQ ID NOs: 12 and 33, or both SEQ ID NOs: 27 and 28. Preferably, the polypeptide comprises the amino acid sequences of both SEQ ID NOs: 11 and 12, both SEQ ID NOs: 33 and 12, or both SEQ ID NOs: 27 and 28.

(45) The invention further provides a protein comprising at least one of the polypeptides described herein. By “protein” is meant a molecule comprising one or more polypeptide chains.

(46) In an embodiment, the protein of the invention can comprise a first polypeptide chain comprising the amino acid sequences of SEQ ID NOs: 3-5, SEQ ID NOs: 13-15, or SEQ ID NOs: 3-4 and 29 and a second polypeptide chain comprising the amino acid sequence of SEQ ID NOs: 6-8, SEQ ID NOs: 16-18, or SEQ ID NOs: 6-7 and 30. Alternatively or additionally, the protein of the invention can comprise a first polypeptide chain comprising the amino acid sequence of SEQ ID NO: 9, 19, or 31 and a second polypeptide chain comprising the amino acid sequence of SEQ ID NO: 10, 20, or 32. The protein of the invention can, for example, comprise a first polypeptide chain comprising the amino acid sequence of SEQ ID NO: 11, 21, 27, or 33 and a second polypeptide chain comprising the amino acid sequence of SEQ ID NO: 12, 22, 28, or 34. In this instance, the protein of the invention can be a TCR. Alternatively, if, for example, the protein comprises a single polypeptide chain comprising SEQ ID NO: 11, 21, 27, or 33 and SEQ ID NO: 12, 22, 28, or 34, or if the first and/or second polypeptide chain(s) of the protein further comprise(s) other amino acid sequences, e.g., an amino acid sequence encoding an immunoglobulin or a portion thereof, then the inventive protein can be a fusion protein. In this regard, the invention also provides a fusion protein comprising at least one of the inventive polypeptides described herein along with at least one other polypeptide. The other polypeptide can exist as a separate polypeptide of the fusion protein, or can exist as a polypeptide, which is expressed in frame (in tandem) with one of the inventive polypeptides described herein. The other polypeptide can encode any peptidic or proteinaceous molecule, or a portion thereof, including, but not limited to an immunoglobulin, CD3, CD4, CD8, an MHC molecule, a CD1 molecule, e.g., CD1a, CD1b, CD1c, CD1d, etc.

(47) The fusion protein can comprise one or more copies of the inventive polypeptide and/or one or more copies of the other polypeptide. For instance, the fusion protein can comprise 1, 2, 3, 4, 5, or more, copies of the inventive polypeptide and/or of the other polypeptide. Suitable methods of making fusion proteins are known in the art, and include, for example, recombinant methods. See, for instance, Choi et al., *Mol. Biotechnol.* 31: 193-202 (2005).

(48) In some embodiments of the invention, the TCRs (and functional portions and functional variants thereof), polypeptides, and proteins of the invention may be expressed as a single protein comprising a linker peptide linking the α chain and the β chain. In this regard, the TCRs (and functional variants and functional portions thereof), polypeptides, and proteins of the invention comprising SEQ ID NO: 11, 21, 27, or 33 and SEQ ID NO: 12, 22, 28, or 34 may further comprise a linker peptide. The linker peptide may advantageously facilitate the expression of a recombinant TCR (including functional portions and functional variants thereof), polypeptide, and/or protein in a host cell. Upon expression of the construct including the linker peptide by a host cell, the linker peptide may be cleaved, resulting in separated α and β chains.

(49) The protein of the invention can be a recombinant antibody comprising at least one of the inventive polypeptides described herein. As used herein, “recombinant antibody” refers to a recombinant (e.g., genetically engineered) protein comprising at least one of the polypeptides of the invention and a polypeptide chain of an antibody, or a portion thereof. The polypeptide of an antibody, or portion thereof, can be a heavy chain, a light chain, a variable or constant region of a

heavy or light chain, a single chain variable fragment (scFv), or an Fc, Fab, or F(ab).sub.2' fragment of an antibody, etc. The polypeptide chain of an antibody, or portion thereof, can exist as a separate polypeptide of the recombinant antibody. Alternatively, the polypeptide chain of an antibody, or portion thereof, can exist as a polypeptide, which is expressed in frame (in tandem) with the polypeptide of the invention. The polypeptide of an antibody, or portion thereof, can be a polypeptide of any antibody or any antibody fragment, including any of the antibodies and antibody fragments described herein.

(50) The TCR (or functional variant thereof), polypeptide, or protein can consist essentially of the specified amino acid sequence or sequences described herein, such that other components of the TCR (or functional variant thereof), polypeptide, or protein, e.g., other amino acids, do not materially change the biological activity of the TCR (or functional variant thereof), polypeptide, or protein. In this regard, the inventive TCR (or functional variant thereof), polypeptide, or protein can, for example, consist essentially of the amino acid sequence of SEQ ID NO: 11, 12, 21, 22, 27, 28, 33, and 34, both SEQ ID NOs: 11 and 12, both SEQ ID NOs: 21 and 22, both SEQ ID NOs: 27 and 28, both SEQ ID NOs: 33 and 34, both SEQ ID NOs: 11 and 34, or both SEQ ID NOs: 12 and 33. Also, for instance, the inventive TCRs (including functional variants thereof), polypeptides, or proteins can consist essentially of the amino acid sequence(s) of SEQ ID NO: 9, 10, 19, 20, 31, 32, both SEQ ID NOs: 9 and 10, both SEQ ID NOs: 19 and 20, both SEQ ID NOs: 31 and 32, both SEQ ID NOs: 9 and 32, both SEQ ID NOs: 10 and 31. Furthermore, the inventive TCRs (including functional variants thereof), polypeptides, or proteins can consist essentially of the amino acid sequence of SEQ ID NO: 3 or 13 (CDR1 of α chain), SEQ ID NO: 4 or 14 (CDR2 of α chain), SEQ ID NO: 5, 15, or 29 (CDR3 of α chain), SEQ ID NO: 6 or 16 (CDR1 of β chain), SEQ ID NO: 7 or 17 (CDR2 of β chain), SEQ ID NO: 8, 18, or 30 (CDR3 of β chain), or any combination thereof, e.g., SEQ ID NOs: 3-5; 6-8; 3-8; 13-15; 16-18; 13-18; 3-4 and 29; 6-7 and 30; 3-4, 29, and 6-8; 3-7 and 30; or 3-4, 29, 6-7, and 30.

(51) The TCRs, polypeptides, and proteins of the invention (including functional variants thereof) can be of any length, i.e., can comprise any number of amino acids, provided that the TCRs, polypeptides, or proteins (or functional variants thereof) retain their biological activity, e.g., the ability to specifically bind to MAGE-A3 and/or MAGE-A6; detect cancer in a mammal; or treat or prevent cancer in a mammal, etc. For example, the polypeptide can be in the range of from about 50 to about 5000 amino acids long, such as 50, 70, 75, 100, 125, 150, 175, 200, 300, 400, 500, 600, 700, 800, 900, 1000 or more amino acids in length. In this regard, the polypeptides of the invention also include oligopeptides.

(52) The TCRs, polypeptides, and proteins of the invention (including functional variants thereof) of the invention can comprise synthetic amino acids in place of one or more naturally-occurring amino acids. Such synthetic amino acids are known in the art, and include, for example, aminocyclohexane carboxylic acid, norleucine, α -amino n-decanoic acid, homoserine, S-acetylaminoethyl-cysteine, trans-3- and trans-4-hydroxyproline, 4-aminophenylalanine, 4-nitrophenylalanine, 4-chlorophenylalanine, 4-carboxyphenylalanine, β -phenylserine β -hydroxyphenylalanine, phenylglycine, α -naphthylalanine, cyclohexylalanine, cyclohexylglycine, indoline-2-carboxylic acid, 1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid, aminomalonic acid, aminomalonic acid monoamide, N'-benzyl-N'-methyl-lysine, N',N'-dibenzyl-lysine, 6-hydroxylysine, ornithine, α -aminocyclopentane carboxylic acid, α -aminocyclohexane carboxylic acid, α -aminocycloheptane carboxylic acid, α -(2-amino-2-norbornane)-carboxylic acid, α , γ -diaminobutyric acid, α , β -diaminopropionic acid, homophenylalanine, and α -tert-butylglycine.

(53) The TCRs, polypeptides, and proteins of the invention (including functional variants thereof) can be glycosylated, amidated, carboxylated, phosphorylated, esterified, N-acylated, cyclized via, e.g., a disulfide bridge, or converted into an acid addition salt and/or optionally dimerized or polymerized, or conjugated.

(54) The TCR, polypeptide, and/or protein of the invention (including functional variants thereof)

can be obtained by methods known in the art. Suitable methods of de novo synthesizing polypeptides and proteins are described in references, such as Chan et al., *Fmoc Solid Phase Peptide Synthesis*, Oxford University Press, Oxford, United Kingdom, 2005; *Peptide and Protein Drug Analysis*, ed. Reid, R., Marcel Dekker, Inc., 2000; *Epitope Mapping*, ed. Westwood et al., Oxford University Press, Oxford, United Kingdom, 2000; and U.S. Pat. No. 5,449,752. Also, polypeptides and proteins can be recombinantly produced using the nucleic acids described herein using standard recombinant methods. See, for instance, Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 3rd ed., Cold Spring Harbor Press, Cold Spring Harbor, NY 2001; and Ausubel et al., *Current Protocols in Molecular Biology*, Greene Publishing Associates and John Wiley & Sons, N Y, 1994. Further, some of the TCRs, polypeptides, and proteins of the invention (including functional variants thereof) can be isolated and/or purified from a source, such as a plant, a bacterium, an insect, a mammal, e.g., a rat, a human, etc. Methods of isolation and purification are well-known in the art. Alternatively, the TCRs, polypeptides, and/or proteins described herein (including functional variants thereof) can be commercially synthesized by companies, such as Synpep (Dublin, CA), Peptide Technologies Corp. (Gaithersburg, MD), and Multiple Peptide Systems (San Diego, CA). In this respect, the inventive TCRs (including functional variants thereof), polypeptides, and proteins can be synthetic, recombinant, isolated, and/or purified.

(55) Included in the scope of the invention are conjugates, e.g., bioconjugates, comprising any of the inventive TCRs, polypeptides, or proteins (including any of the functional variants thereof), nucleic acids, recombinant expression vectors, host cells, populations of host cells, or antibodies, or antigen binding portions thereof. Conjugates, as well as methods of synthesizing conjugates in general, are known in the art (See, for instance, Hudecz, F., *Methods Mol. Biol.* 298: 209-223 (2005) and Kirin et al., *Inorg Chem.* 44(15): 5405-5415 (2005)).

(56) By “nucleic acid” as used herein includes “polynucleotide,” “oligonucleotide,” and “nucleic acid molecule,” and generally means a polymer of DNA or RNA, which can be single-stranded or double-stranded, synthesized or obtained (e.g., isolated and/or purified) from natural sources, which can contain natural, non-natural or altered nucleotides, and which can contain a natural, non-natural or altered internucleotide linkage, such as a phosphoroamidate linkage or a phosphorothioate linkage, instead of the phosphodiester found between the nucleotides of an unmodified oligonucleotide. It is generally preferred that the nucleic acid does not comprise any insertions, deletions, inversions, and/or substitutions. However, it may be suitable in some instances, as discussed herein, for the nucleic acid to comprise one or more insertions, deletions, inversions, and/or substitutions.

(57) Preferably, the nucleic acids of the invention are recombinant. As used herein, the term “recombinant” refers to (i) molecules that are constructed outside living cells by joining natural or synthetic nucleic acid segments to nucleic acid molecules that can replicate in a living cell, or (ii) molecules that result from the replication of those described in (i) above. For purposes herein, the replication can be in vitro replication or in vivo replication.

(58) The nucleic acids can be constructed based on chemical synthesis and/or enzymatic ligation reactions using procedures known in the art. See, for example, Sambrook et al., *supra*, and Ausubel et al., *supra*. For example, a nucleic acid can be chemically synthesized using naturally occurring nucleotides or variously modified nucleotides designed to increase the biological stability of the molecules or to increase the physical stability of the duplex formed upon hybridization (e.g., phosphorothioate derivatives and acridine substituted nucleotides). Examples of modified nucleotides that can be used to generate the nucleic acids include, but are not limited to, 5-fluorouracil, 5-bromouracil, 5-chlorouracil, 5-iodouracil, hypoxanthine, xanthine, 4-acetylcytosine, 5-(carboxyhydroxymethyl) uracil, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluracil, dihydrouracil, beta-D-galactosylqueosine, inosine, N⁶-isopentenyladenine, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-

methylguanine, 3-methylcytosine, 5-methylcytosine, N.sup.6-substituted adenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5'-methoxycarboxymethyluracil, 5-methoxyuracil, 2-methylthio-N.sup.6-isopentenyladenine, uracil-5-oxyacetic acid (v), wybutoxosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, uracil-5-oxyacetic acid methylester, 3-(3-amino-3-N-2-carboxypropyl) uracil, and 2,6-diaminopurine. Alternatively, one or more of the nucleic acids of the invention can be purchased from companies, such as Macromolecular Resources (Fort Collins, CO) and Synthesgen (Houston, TX).

(59) The nucleic acid can comprise any nucleotide sequence which encodes any of the TCRs, polypeptides, proteins, or functional functional variants thereof described herein. For example, the nucleic acid can comprise, consist, or consist essentially of any one or more of the nucleotide sequence SEQ ID NOs: 37-44.

(60) The invention also provides a nucleic acid comprising a nucleotide sequence which is complementary to the nucleotide sequence of any of the nucleic acids described herein or a nucleotide sequence which hybridizes under stringent conditions to the nucleotide sequence of any of the nucleic acids described herein.

(61) The nucleotide sequence which hybridizes under stringent conditions preferably hybridizes under high stringency conditions. By "high stringency conditions" is meant that the nucleotide sequence specifically hybridizes to a target sequence (the nucleotide sequence of any of the nucleic acids described herein) in an amount that is detectably stronger than non-specific hybridization. High stringency conditions include conditions which would distinguish a polynucleotide with an exact complementary sequence, or one containing only a few scattered mismatches from a random sequence that happened to have a few small regions (e.g., 3-10 bases) that matched the nucleotide sequence. Such small regions of complementarity are more easily melted than a full-length complement of 14-17 or more bases, and high stringency hybridization makes them easily distinguishable. Relatively high stringency conditions would include, for example, low salt and/or high temperature conditions, such as provided by about 0.02-0.1 M NaCl or the equivalent, at temperatures of about 50-70° C. Such high stringency conditions tolerate little, if any, mismatch between the nucleotide sequence and the template or target strand, and are particularly suitable for detecting expression of any of the inventive TCRs (including functional portions and functional variants thereof). It is generally appreciated that conditions can be rendered more stringent by the addition of increasing amounts of formamide.

(62) The invention also provides a nucleic acid comprising a nucleotide sequence that is at least about 70% or more, e.g., about 80%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% identical to any of the nucleic acids described herein.

(63) The nucleic acids of the invention can be incorporated into a recombinant expression vector. In this regard, the invention provides recombinant expression vectors comprising any of the nucleic acids of the invention. For purposes herein, the term "recombinant expression vector" means a genetically-modified oligonucleotide or polynucleotide construct that permits the expression of an mRNA, protein, polypeptide, or peptide by a host cell, when the construct comprises a nucleotide sequence encoding the mRNA, protein, polypeptide, or peptide, and the vector is contacted with the cell under conditions sufficient to have the mRNA, protein, polypeptide, or peptide expressed within the cell. The vectors of the invention are not naturally-occurring as a whole. However, parts of the vectors can be naturally-occurring. The inventive recombinant expression vectors can comprise any type of nucleotides, including, but not limited to DNA and RNA, which can be single-stranded or double-stranded, synthesized or obtained in part from natural sources, and which can contain natural, non-natural or altered nucleotides. The recombinant expression vectors can comprise naturally-occurring, non-naturally-occurring internucleotide linkages, or both types of linkages. Preferably, the non-naturally occurring or altered nucleotides or internucleotide linkages

does not hinder the transcription or replication of the vector.

(64) The recombinant expression vector of the invention can be any suitable recombinant expression vector, and can be used to transform or transfect any suitable host cell. Suitable vectors include those designed for propagation and expansion or for expression or both, such as plasmids and viruses. The vector can be selected from the group consisting of the pUC series (Fermentas Life Sciences), the pBluescript series (Stratagene, LaJolla, CA), the pET series (Novagen, Madison, WI), the pGEX series (Pharmacia Biotech, Uppsala, Sweden), and the pEX series (Clontech, Palo Alto, CA). Bacteriophage vectors, such as λ GT10, λ GT11, λ ZapII (Stratagene), λ EMBL4, and λ NM1149, also can be used. Examples of plant expression vectors include pBI01, pBI101.2, pBI101.3, pBI121 and pBIN19 (Clontech). Examples of animal expression vectors include pEUK-Cl, pMAM and pMAMneo (Clontech). Preferably, the recombinant expression vector is a viral vector, e.g., a retroviral vector.

(65) The recombinant expression vectors of the invention can be prepared using standard recombinant DNA techniques described in, for example, Sambrook et al., supra, and Ausubel et al., supra. Constructs of expression vectors, which are circular or linear, can be prepared to contain a replication system functional in a prokaryotic or eukaryotic host cell. Replication systems can be derived, e.g., from ColEI, 2 μ plasmid, λ , SV40, bovine papilloma virus, and the like.

(66) Desirably, the recombinant expression vector comprises regulatory sequences, such as transcription and translation initiation and termination codons, which are specific to the type of host cell (e.g., bacterium, fungus, plant, or animal) into which the vector is to be introduced, as appropriate and taking into consideration whether the vector is DNA- or RNA-based.

(67) The recombinant expression vector can include one or more marker genes, which allow for selection of transformed or transfected host cells. Marker genes include biocide resistance, e.g., resistance to antibiotics, heavy metals, etc., complementation in an auxotrophic host cell to provide prototrophy, and the like. Suitable marker genes for the inventive expression vectors include, for instance, neomycin/G418 resistance genes, hygromycin resistance genes, histidinol resistance genes, tetracycline resistance genes, and ampicillin resistance genes.

(68) The recombinant expression vector can comprise a native or nonnative promoter operably linked to the nucleotide sequence encoding the TCR, polypeptide, or protein (including functional variants thereof), or to the nucleotide sequence which is complementary to or which hybridizes to the nucleotide sequence encoding the TCR, polypeptide, or protein (including functional variants thereof). The selection of promoters, e.g., strong, weak, inducible, tissue-specific and developmental-specific, is within the ordinary skill of the artisan. Similarly, the combining of a nucleotide sequence with a promoter is also within the skill of the artisan. The promoter can be a non-viral promoter or a viral promoter, e.g., a cytomegalovirus (CMV) promoter, an SV40 promoter, an RSV promoter, and a promoter found in the long-terminal repeat of the murine stem cell virus.

(69) The inventive recombinant expression vectors can be designed for either transient expression, for stable expression, or for both. Also, the recombinant expression vectors can be made for constitutive expression or for inducible expression. Further, the recombinant expression vectors can be made to include a suicide gene.

(70) As used herein, the term "suicide gene" refers to a gene that causes the cell expressing the suicide gene to die. The suicide gene can be a gene that confers sensitivity to an agent, e.g., a drug, upon the cell in which the gene is expressed, and causes the cell to die when the cell is contacted with or exposed to the agent. Suicide genes are known in the art (see, for example, *Suicide Gene Therapy: Methods and Reviews*, Springer, Caroline J. (Cancer Research UK Centre for Cancer Therapeutics at the Institute of Cancer Research, Sutton, Surrey, UK), Humana Press, 2004) and include, for example, the Herpes Simplex Virus (HSV) thymidine kinase (TK) gene, cytosine deaminase, purine nucleoside phosphorylase, and nitroreductase.

(71) Another embodiment of the invention further provides a host cell comprising any of the

recombinant expression vectors described herein. As used herein, the term “host cell” refers to any type of cell that can contain the inventive recombinant expression vector. The host cell can be a eukaryotic cell, e.g., plant, animal, fungi, or algae, or can be a prokaryotic cell, e.g., bacteria or protozoa. The host cell can be a cultured cell or a primary cell, i.e., isolated directly from an organism, e.g., a human. The host cell can be an adherent cell or a suspended cell, i.e., a cell that grows in suspension. Suitable host cells are known in the art and include, for instance, DH5 α *E. coli* cells, Chinese hamster ovarian cells, monkey VERO cells, COS cells, HEK293 cells, and the like. For purposes of amplifying or replicating the recombinant expression vector, the host cell is preferably a prokaryotic cell, e.g., a DH5 α cell. For purposes of producing a recombinant TCR, polypeptide, or protein, the host cell is preferably a mammalian cell. Most preferably, the host cell is a human cell. While the host cell can be of any cell type, can originate from any type of tissue, and can be of any developmental stage, the host cell preferably is a peripheral blood lymphocyte (PBL) or a peripheral blood mononuclear cell (PBMC). More preferably, the host cell is a T cell. (72) For purposes herein, the T cell can be any T cell, such as a cultured T cell, e.g., a primary T cell, or a T cell from a cultured T cell line, e.g., Jurkat, SupT1, etc., or a T cell obtained from a mammal. If obtained from a mammal, the T cell can be obtained from numerous sources, including but not limited to blood, bone marrow, lymph node, the thymus, or other tissues or fluids. T cells can also be enriched for or purified. Preferably, the T cell is a human T cell. More preferably, the T cell is a T cell isolated from a human. The T cell can be any type of T cell and can be of any developmental stage, including but not limited to, CD4^{sup.}+ / CD8^{sup.}+ double positive T cells, CD4^{sup.}+ helper T cells, e.g., Th.sub.1 and Th.sub.2 cells, CD4⁺ T cells, CD8^{sup.}+ T cells (e.g., cytotoxic T cells), tumor infiltrating lymphocytes (TILs), memory T cells (e.g., central memory T cells and effector memory T cells), naïve T cells, and the like.

(73) Also provided by the invention is a population of cells comprising at least one host cell described herein. The population of cells can be a heterogeneous population comprising the host cell comprising any of the recombinant expression vectors described, in addition to at least one other cell, e.g., a host cell (e.g., a T cell), which does not comprise any of the recombinant expression vectors, or a cell other than a T cell, e.g., a B cell, a macrophage, a neutrophil, an erythrocyte, a hepatocyte, an endothelial cell, an epithelial cells, a muscle cell, a brain cell, etc. Alternatively, the population of cells can be a substantially homogeneous population, in which the population comprises mainly of host cells (e.g., consisting essentially of) comprising the recombinant expression vector. The population also can be a clonal population of cells, in which all cells of the population are clones of a single host cell comprising a recombinant expression vector, such that all cells of the population comprise the recombinant expression vector. In one embodiment of the invention, the population of cells is a clonal population comprising host cells comprising a recombinant expression vector as described herein.

(74) The invention further provides an antibody, or antigen binding portion thereof, which specifically binds to a functional portion of any of the TCRs (or functional variant thereof) described herein. Preferably, the functional portion specifically binds to the cancer antigen, e.g., the functional portion comprising the amino acid sequence SEQ ID NO: 3 or 13 (CDR1 of α chain), 4 or 14 (CDR2 of α chain), 5, 15, or 29 (CDR3 of α chain), 6 or 16 (CDR1 of β chain), 7 or 17 (CDR2 of β chain), 8, 18, or 30 (CDR3 of β chain), SEQ ID NO: 9, 19, or 31 (variable region of α chain), SEQ ID NO: 10, 20, or 32 (variable region of β chain), or a combination thereof, e.g., 3-5; 6-8; 3-8; 13-15; 16-18; 13-18; 3-4 and 29; 6-7 and 30; 3-4, 29, and 6-8; or 3-7 and 30; 3-4, 29, 6-7, and 30. More preferably, the functional portion comprises the amino acid sequences of SEQ ID NOs: 3-8 or SEQ ID NOs: 3-4, 29, and 6-8. In a preferred embodiment, the antibody, or antigen binding portion thereof, binds to an epitope which is formed by all 6 CDRs (CDR1-3 of the alpha chain and CDR1-3 of the beta chain). The antibody can be any type of immunoglobulin that is known in the art. For instance, the antibody can be of any isotype, e.g., IgA, IgD, IgE, IgG, IgM, etc. The antibody can be monoclonal or polyclonal. The antibody can be a naturally-occurring

antibody, e.g., an antibody isolated and/or purified from a mammal, e.g., mouse, rabbit, goat, horse, chicken, hamster, human, etc. Alternatively, the antibody can be a genetically-engineered antibody, e.g., a humanized antibody or a chimeric antibody. The antibody can be in monomeric or polymeric form. Also, the antibody can have any level of affinity or avidity for the functional portion of the inventive TCR (or functional variant thereof). Desirably, the antibody is specific for the functional portion of the inventive TCR (or functional variants thereof), such that there is minimal cross-reaction with other peptides or proteins.

(75) Methods of testing antibodies for the ability to bind to any functional portion or functional variant of the inventive TCR are known in the art and include any antibody-antigen binding assay, such as, for example, radioimmunoassay (RIA), ELISA, Western blot, immunoprecipitation, and competitive inhibition assays (see, e.g., Janeway et al., *infra*, and U.S. Patent Application Publication No. 2002/0197266 A1).

(76) Suitable methods of making antibodies are known in the art. For instance, standard hybridoma methods are described in, e.g., Kohler and Milstein, *Eur. J. Immunol.*, 5, 511-519 (1976), Harlow and Lane (eds.), *Antibodies: A Laboratory Manual*, CSH Press (1988), and C. A. Janeway et al. (eds.), *Immunobiology*, 5^{sup.th} Ed., Garland Publishing, New York, NY (2001)). Alternatively, other methods, such as EBV-hybridoma methods (Haskard and Archer, *J. Immunol. Methods*, 74(2), 361-67 (1984), and Roder et al., *Methods Enzymol.*, 121, 140-67 (1986)), and bacteriophage vector expression systems (see, e.g., Huse et al., *Science*, 246, 1275-81 (1989)) are known in the art. Further, methods of producing antibodies in non-human animals are described in, e.g., U.S. Pat. Nos. 5,545,806, 5,569,825, and 5,714,352, and U.S. Patent Application Publication No. 2002/0197266 A1.

(77) Phage display furthermore can be used to generate the antibody of the invention. In this regard, phage libraries encoding antigen-binding variable (V) domains of antibodies can be generated using standard molecular biology and recombinant DNA techniques (see, e.g., Sambrook et al. (eds.), *Molecular Cloning, A Laboratory Manual*, 3^{sup.rd} Edition, Cold Spring Harbor Laboratory Press, New York (2001)). Phage encoding a variable region with the desired specificity are selected for specific binding to the desired antigen, and a complete or partial antibody is reconstituted comprising the selected variable domain. Nucleic acid sequences encoding the reconstituted antibody are introduced into a suitable cell line, such as a myeloma cell used for hybridoma production, such that antibodies having the characteristics of monoclonal antibodies are secreted by the cell (see, e.g., Janeway et al., *supra*, Huse et al., *supra*, and U.S. Pat. No. 6,265,150).

(78) Antibodies can be produced by transgenic mice that are transgenic for specific heavy and light chain immunoglobulin genes. Such methods are known in the art and described in, for example U.S. Pat. Nos. 5,545,806 and 5,569,825, and Janeway et al., *supra*.

(79) Methods for generating humanized antibodies are well known in the art and are described in detail in, for example, Janeway et al., *supra*, U.S. Pat. Nos. 5,225,539, 5,585,089 and 5,693,761, European Patent No. 0239400 B1, and United Kingdom Patent No. 2188638. Humanized antibodies can also be generated using the antibody resurfacing technology described in, for example, U.S. Pat. No. 5,639,641 and Pedersen et al., *J. Mol. Biol.*, 235, 959-973 (1994).

(80) The invention also provides antigen binding portions of any of the antibodies described herein. The antigen binding portion can be any portion that has at least one antigen binding site, such as Fab, F(ab').sub.2, dsFv, sFv, diabodies, and triabodies.

(81) A single-chain variable region fragment (sFv) antibody fragment, which consists of a truncated Fab fragment comprising the variable (V) domain of an antibody heavy chain linked to a V domain of a light antibody chain via a synthetic peptide, can be generated using routine recombinant DNA technology techniques (see, e.g., Janeway et al., *supra*). Similarly, disulfide-stabilized variable region fragments (dsFv) can be prepared by recombinant DNA technology (see, e.g., Reiter et al., *Protein Engineering*, 7, 697-704 (1994)). Antibody fragments of the invention, however, are not

limited to these exemplary types of antibody fragments.

(82) Also, the antibody, or antigen binding portion thereof, can be modified to comprise a detectable label, such as, for instance, a radioisotope, a fluorophore (e.g., fluorescein isothiocyanate (FITC), phycoerythrin (PE)), an enzyme (e.g., alkaline phosphatase, horseradish peroxidase), and element particles (e.g., gold particles).

(83) The inventive TCRs, polypeptides, proteins, (including functional variants thereof), nucleic acids, recombinant expression vectors, host cells (including populations thereof), and antibodies (including antigen binding portions thereof), can be isolated and/or purified. The term “isolated” as used herein means having been removed from its natural environment. The term “purified” as used herein means having been increased in purity, wherein “purity” is a relative term, and not to be necessarily construed as absolute purity. For example, the purity can be at least about 50%, can be greater than 60%, 70%, 80%, 90%, 95%, or can be 100%.

(84) The inventive TCRs, polypeptides, proteins (including functional variants thereof), nucleic acids, recombinant expression vectors, host cells (including populations thereof), and antibodies (including antigen binding portions thereof), all of which are collectively referred to as “inventive TCR materials” hereinafter, can be formulated into a composition, such as a pharmaceutical composition. In this regard, the invention provides a pharmaceutical composition comprising any of the TCRs, polypeptides, proteins, functional portions, functional variants, nucleic acids, expression vectors, host cells (including populations thereof), and antibodies (including antigen binding portions thereof), and a pharmaceutically acceptable carrier. The inventive pharmaceutical compositions containing any of the inventive TCR materials can comprise more than one inventive TCR material, e.g., a polypeptide and a nucleic acid, or two or more different TCRs (including functional portions and functional variants thereof). Alternatively, the pharmaceutical composition can comprise an inventive TCR material in combination with another pharmaceutically active agents or drugs, such as a chemotherapeutic agents, e.g., asparaginase, busulfan, carboplatin, cisplatin, daunorubicin, doxorubicin, fluorouracil, gemcitabine, hydroxyurea, methotrexate, paclitaxel, rituximab, vinblastine, vincristine, etc.

(85) Preferably, the carrier is a pharmaceutically acceptable carrier. With respect to pharmaceutical compositions, the carrier can be any of those conventionally used for the particular inventive TCR material under consideration. Such pharmaceutically acceptable carriers are well-known to those skilled in the art and are readily available to the public. It is preferred that the pharmaceutically acceptable carrier be one which has no detrimental side effects or toxicity under the conditions of use.

(86) The choice of carrier will be determined in part by the particular inventive TCR material, as well as by the particular method used to administer the inventive TCR material. Accordingly, there are a variety of suitable formulations of the pharmaceutical composition of the invention. Suitable formulations may include any of those for oral, aerosol, parenteral, subcutaneous, intravenous, intramuscular, intraarterial, intrathecal, or interperitoneal administration. More than one route can be used to administer the inventive TCR materials, and in certain instances, a particular route can provide a more immediate and more effective response than another route.

(87) Preferably, the inventive TCR material is administered by injection, e.g., intravenously. When the inventive TCR material is a host cell expressing the inventive TCR (or functional variant thereof), the pharmaceutically acceptable carrier for the cells for injection may include any isotonic carrier such as, for example, normal saline (about 0.90% w/v of NaCl in water, about 300 mOsm/L NaCl in water, or about 9.0 g NaCl per liter of water), NORMOSOL R electrolyte solution (Abbott, Chicago, IL), PLASMA-LYTE A (Baxter, Deerfield, IL), about 5% dextrose in water, or Ringer's lactate. In an embodiment, the pharmaceutically acceptable carrier is supplemented with human serum albumen.

(88) In an embodiment of the invention, the pharmaceutical composition may further comprise MHC Class I restricted TCRs, or polypeptides, proteins, nucleic acids, or recombinant expression

vectors encoding MHC Class I restricted TCRs, or host cells or populations of cells expressing MHC Class I restricted TCRs. Without being bound to a particular theory, it is believed that MHC Class I restricted CD8⁺ T cells augment the reactivity of MHC Class II restricted CD4⁺ T cells and enhance the ability of the MHC Class II restricted CD4⁺ T cells to treat or prevent cancer.

(89) For purposes of the invention, the amount or dose (e.g., numbers of cells when the inventive TCR material is one or more cells) of the inventive TCR material administered should be sufficient to effect, e.g., a therapeutic or prophylactic response, in the subject or animal over a reasonable time frame. For example, the dose of the inventive TCR material should be sufficient to bind to a cancer antigen, or detect, treat or prevent cancer in a period of from about 2 hours or longer, e.g., 12 to 24 or more hours, from the time of administration. In certain embodiments, the time period could be even longer. The dose will be determined by the efficacy of the particular inventive TCR material and the condition of the animal (e.g., human), as well as the body weight of the animal (e.g., human) to be treated.

(90) Many assays for determining an administered dose are known in the art. For purposes of the invention, an assay, which comprises comparing the extent to which target cells are lysed or IFN- γ is secreted by T cells expressing the inventive TCR (or functional variant or functional portion thereof), polypeptide, or protein upon administration of a given dose of such T cells to a mammal among a set of mammals of which is each given a different dose of the T cells, could be used to determine a starting dose to be administered to a mammal. The extent to which target cells are lysed or IFN- γ is secreted upon administration of a certain dose can be assayed by methods known in the art.

(91) The dose of the inventive TCR material also will be determined by the existence, nature and extent of any adverse side effects that might accompany the administration of a particular inventive TCR material. Typically, the attending physician will decide the dosage of the inventive TCR material with which to treat each individual patient, taking into consideration a variety of factors, such as age, body weight, general health, diet, sex, inventive TCR material to be administered, route of administration, and the severity of the condition being treated. In an embodiment in which the inventive TCR material is a population of cells, the number of cells administered per infusion may vary, e.g., from about 1×10^6 to about 1×10^{11} cells or more.

(92) One of ordinary skill in the art will readily appreciate that the inventive TCR materials of the invention can be modified in any number of ways, such that the therapeutic or prophylactic efficacy of the inventive TCR materials is increased through the modification. For instance, the inventive TCR materials can be conjugated either directly or indirectly through a bridge to a targeting moiety. The practice of conjugating compounds, e.g., inventive TCR materials, to targeting moieties is known in the art. See, for instance, Wadwa et al., *J Drug Targeting* 3: 111 (1995) and U.S. Pat. No. 5,087,616. The term “targeting moiety” as used herein, refers to any molecule or agent that specifically recognizes and binds to a cell-surface receptor, such that the targeting moiety directs the delivery of the inventive TCR materials to a population of cells on which surface the receptor is expressed. Targeting moieties include, but are not limited to, antibodies, or fragments thereof, peptides, hormones, growth factors, cytokines, and any other natural or non-natural ligands, which bind to cell surface receptors (e.g., Epithelial Growth Factor Receptor (EGFR), T-cell receptor (TCR), B-cell receptor (BCR), CD28, Platelet-derived Growth Factor Receptor (PDGF), nicotinic acetylcholine receptor (nAChR), etc.). The term “bridge” as used herein, refers to any agent or molecule that links the inventive TCR materials to the targeting moiety. One of ordinary skill in the art recognizes that sites on the inventive TCR materials, which are not necessary for the function of the inventive TCR materials, are ideal sites for attaching a bridge and/or a targeting moiety, provided that the bridge and/or targeting moiety, once attached to the inventive TCR materials, do(es) not interfere with the function of the inventive TCR materials, i.e., the ability to bind to MAGE-A3 or MAGE-A6; or to detect, treat, or prevent cancer.

(93) It is contemplated that the inventive pharmaceutical compositions, TCRs (including functional

variants thereof), polypeptides, proteins, nucleic acids, recombinant expression vectors, host cells, or populations of cells can be used in methods of treating or preventing cancer. Without being bound to a particular theory, the inventive TCRs (and functional variants thereof) are believed to bind specifically to MAGE-A3 and/or MAGE-A6, such that the TCR (or related inventive polypeptide or protein and functional variants thereof) when expressed by a cell is able to mediate an immune response against a target cell expressing MAGE-A3 or MAGE-A6. In this regard, the invention provides a method of treating or preventing cancer in a mammal, comprising administering to the mammal any of the pharmaceutical compositions, TCRs (and functional variants thereof), polypeptides, or proteins described herein, any nucleic acid or recombinant expression vector comprising a nucleotide sequence encoding any of the TCRs (and functional variants thereof), polypeptides, proteins described herein, or any host cell or population of cells comprising a recombinant vector which encodes any of the TCRs (and functional variants thereof), polypeptides, or proteins described herein, in an amount effective to treat or prevent cancer in the mammal.

(94) In an embodiment of the invention, the inventive methods of treating or preventing cancer may further comprise co-administering MHC Class I restricted TCRs, or polypeptides, proteins, nucleic acids, or recombinant expression vectors encoding MHC Class I restricted TCRs, or host cells or populations of cells expressing MHC Class I restricted TCRs, to the mammal.

(95) The terms “treat,” and “prevent” as well as words stemming therefrom, as used herein, do not necessarily imply 100% or complete treatment or prevention. Rather, there are varying degrees of treatment or prevention of which one of ordinary skill in the art recognizes as having a potential benefit or therapeutic effect. In this respect, the inventive methods can provide any amount of any level of treatment or prevention of cancer in a mammal. Furthermore, the treatment or prevention provided by the inventive method can include treatment or prevention of one or more conditions or symptoms of the disease, e.g., cancer, being treated or prevented. Also, for purposes herein, “prevention” can encompass delaying the onset of the disease, or a symptom or condition thereof.

(96) Also provided is a method of detecting the presence of cancer in a mammal. The method comprises (i) contacting a sample comprising cells of the cancer with any of the inventive TCRs (and functional variants thereof), polypeptides, proteins, nucleic acids, recombinant expression vectors, host cells, populations of cells, or antibodies, or antigen binding portions thereof, described herein, thereby forming a complex, and detecting the complex, wherein detection of the complex is indicative of the presence of cancer in the mammal.

(97) With respect to the inventive method of detecting cancer in a mammal, the sample of cells of the cancer can be a sample comprising whole cells, lysates thereof, or a fraction of the whole cell lysates, e.g., a nuclear or cytoplasmic fraction, a whole protein fraction, or a nucleic acid fraction.

(98) For purposes of the inventive detecting method, the contacting can take place in vitro or in vivo with respect to the mammal. Preferably, the contacting is in vitro.

(99) Also, detection of the complex can occur through any number of ways known in the art. For instance, the inventive TCRs (and functional variants thereof), polypeptides, proteins, nucleic acids, recombinant expression vectors, host cells, populations of cells, or antibodies, or antigen binding portions thereof, described herein, can be labeled with a detectable label such as, for instance, a radioisotope, a fluorophore (e.g., fluorescein isothiocyanate (FITC), phycoerythrin (PE)), an enzyme (e.g., alkaline phosphatase, horseradish peroxidase), and element particles (e.g., gold particles).

(100) For purposes of the inventive methods, wherein host cells or populations of cells are administered, the cells can be cells that are allogeneic or autologous to the mammal. Preferably, the cells are autologous to the mammal.

(101) With respect to the inventive methods, the cancer can be any cancer, including any of sarcomas (e.g., synovial sarcoma, osteogenic sarcoma, leiomyosarcoma uteri, and alveolar rhabdomyosarcoma), lymphomas (e.g., Hodgkin lymphoma and non-Hodgkin lymphoma),

hepatocellular carcinoma, glioma, head cancers (e.g., squamous cell carcinoma), neck cancers (e.g., squamous cell carcinoma), acute lymphocytic cancer, leukemias (e.g., acute myeloid leukemia and chronic lymphocytic leukemia), bone cancer, brain cancer, breast cancer, cancer of the anus, anal canal, or anorectum, cancer of the eye, cancer of the intrahepatic bile duct, cancer of the joints, cancer of the neck, gallbladder, or pleura, cancer of the nose, nasal cavity, or middle ear, cancer of the oral cavity, cancer of the vulva, chronic myeloid cancer, colon cancers (e.g., colon carcinoma), esophageal cancer, cervical cancer, gastric cancer, gastrointestinal carcinoid tumor, hypopharynx cancer, larynx cancer, liver cancers (e.g., hepatocellular carcinoma), lung cancers (e.g., non-small cell lung carcinoma), malignant mesothelioma, melanoma, multiple myeloma, nasopharynx cancer, ovarian cancer, pancreatic cancer, peritoneum, omentum, and mesentery cancer, pharynx cancer, prostate cancer, rectal cancer, kidney cancers (e.g., renal cell carcinoma), small intestine cancer, soft tissue cancer, stomach cancer, testicular cancer, thyroid cancer, and urothelial cancers (e.g., ureter cancer and urinary bladder cancer). Preferably, the cancer is melanoma, breast cancer, lung cancer, prostate cancer, synovial cell sarcoma, head and neck cancer, esophageal cancer, or ovarian cancer.

(102) The mammal referred to in the inventive methods can be any mammal. As used herein, the term “mammal” refers to any mammal, including, but not limited to, mammals of the order Rodentia, such as mice and hamsters, and mammals of the order Logomorpha, such as rabbits. It is preferred that the mammals are from the order Carnivora, including Felines (cats) and Canines (dogs). It is more preferred that the mammals are from the order Artiodactyla, including Bovines (cows) and Swines (pigs) or of the order Perssodactyla, including Equines (horses). It is most preferred that the mammals are of the order Primates, Ceboids, or Simoids (monkeys) or of the order Anthropoids (humans and apes). An especially preferred mammal is the human.

(103) The following examples further illustrate the invention but, of course, should not be construed as in any way limiting its scope.

EXAMPLE 1A

(104) This example demonstrates the isolation of TCRs from T cell clones.

(105) Anti-MAGE-A3.sub.243-258 CD4⁺ effector clone R12C9 and anti-MAGE-A3.sub.243-258 Treg clone 6F9 was cultured with peptide (MAGE-A3.sub.243-258)-pulsed EBV B cells. Cytokine secretion, percentage of indicator cells suppressed, percentage of FOXP3⁺ Treg cells, and percentage of unmethylated FOXP3 sequences were measured. Unmethylated FOXP3 intron 1 sequences are considered to be a marker for a stable Treg phenotype. The results for the 6F9 and R12C9 clones are shown in Tables 1A and 1B.

(106) TABLE-US-00001 TABLE 1A % Indicator Cells % Unmethylated Clone Suppressed % FOXP3⁺ FOXP3 sequences 6F9 57 95 72 R12C9 0 8 2

(107) TABLE-US-00002 TABLE 1B Cytokine Secretion (pg/25,000 cells) Clone IFN- γ IL-2 IL-10 IL-4 IL-5 TNF- α 6F9 0 0 0 0 0 12 R12C9 922 46 479 8 28 422

(108) Treg clones can inhibit the proliferation of indicator cells after stimulation by an appropriate peptide. As shown in Table 1A, clone 6F9 is a Treg clone.

(109) A TCR comprising SEQ ID NOs: 21 and 22 was cloned from the anti-MAGE-A3.sub.243-258 CD4⁺ effector clone R12C9 (“R12C9 TCR”). A TCR comprising SEQ ID NOs: 11 and 12 was cloned from the anti-MAGE-A3.sub.243-258 Treg clone 6F9 (“6F9 TCR”).

EXAMPLE 1B

(110) This example demonstrates the transduction efficiency of PBMC transduced with a nucleotide sequence encoding the 6F9 TCR or R12C9 TCR of Example 1.

(111) Transcripts encoding the TCR alpha and beta chains of R12C9 and 6F9 were linked with sequences encoding a P2A self-cleaving peptide and cloned into an MSGV1 retroviral vector. PBMC from three patients were stimulated with OKT3, transduced with transient retroviral supernatants on day two, and enriched for CD4⁺ T cells on day seven. The levels of TCR expression were evaluated by staining transduced cells with anti-V β 22 or V β 6.7, which detect the

6F9 or R12C9 TCR, respectively. Analysis of PBMC from patient 1 indicated that between 25 and 35% of the T cells were transduced with the individual TCRs and similar transduction levels were obtained with PBMC from patients 2 and 3.

EXAMPLE 2

(112) This example demonstrates that T cells transduced with nucleotide sequences encoding the anti-MAGE-A3.sub.243-258 TCRs of Example 1 recognize 293-class II, major histocompatibility complex, transactivator (CIITA) transfectants of MAGE-A3 and peptide-pulsed targets. This example also demonstrates that the 6F9 TCR recognizes 293-CIITA transfectants of MAGE-A3 and MAGE-A6.

(113) CD4⁺ enriched peripheral blood lymphocytes (PBL) from two human donors were untransduced (UT) or transduced with F5 (anti-MART-1) TCR, R12C9 TCR, or 6F9 TCR. The cells were cultured with 293-CIITA-transfected target cells pulsed with MAGE-A3.sub.243-258 (SEQ ID NO: 2) peptide. The 293-CIITA cells are 293 cells transduced with CIITA, which is a human gene which encodes the class II, major histocompatibility complex transactivator protein. The results obtained with 6F9 and R12C9 TCR-transduced cells are shown in Table 2 and FIG.

10A. PBL transduced by R12C9 TCR or 6F9 TCR recognized MAGE-A3.sub.243-258 peptide-pulsed HLA-DP*0401⁺ target cells. Titration of the MAGE-A3.sub.243-258 peptide indicated that CD4^{sup.}+ T cells transduced with the 6F9 or R12C9 TCRs released comparable levels of IFN- γ in response to targets pulsed with a minimum of between 0.001 and 0.01 mg/ml of the MAGE-A3:sub.243-258 peptide. The experiments were repeated using PBL from a third human donor and similar results were obtained.

(114) TABLE-US-00003

TABLE 2 Donor 1 IFN-gamma (pg/ml)		Peptide		Untransduced		6F9 TCR	
(μ g/ml)	(UT)	transduced	293-CIITA + MAGE-	DP4+	0.0001	2	328
A3.sub.243-258	0.001	17	596	0.01	4	1609	0.1
2	7440	1	4	34800	10	0	52100

Donor 2 IFN-gamma (pg/ml)

Peptide		Untransduced		6F9 TCR	
(μ g/ml)	(UT)	transduced	293-CIITA + MAGE-	DP4+	0.0001
28	110	A3.sub.243-258	0.001	30	323
0.01	37	1830	0.1	40	9760
1	44	55000	10	0	59050

(115) 293-CIITA target cells were transfected with DNA constructs (pCDNA3 vector) encoding full-length MAGE-A3 protein or MAGE-A6 protein, which differ at only a single position (249), or full-length MAGE-A1 protein or MAGE-A12 protein. Untransduced and transduced PBL were co-cultured with the transfected 293-CIITA cells and interferon (IFN) gamma secretion was measured. The results are shown in FIGS. 1A, 1B, and 10A.

(116) As shown in FIGS. 1A, 1B, and 10A, although T cells transduced with R12C9 TCR or 6F9 TCR recognized peptide-pulsed targets, PBL transduced with the 6F9 TCR were the most highly reactive to each of MAGE-A3 and MAGE-A6 293-CIITA transfectants. CD4⁺ T cells transduced with the 6F9 but not the R12C9 TCR recognized HLA DP*0401.sup.+293-CIITA cells transfected with genes encoding MAGE-A3 or MAGE-A6, but not MAGE-A1 or A12. Comparison of amino acid sequences of the corresponding regions of the MAGE family members indicated that MAGE-A3 and MAGE-A6 only differed at one position (residue 249), whereas the other MAGE family members differed from MAGE-A3 at two (MAGE-A12.sub.243-258 (SEQ ID NO: 70)) or three (MAGE-A1.sub.243-258 (SEQ ID NO: 71)) positions. In addition, CD4^{sup.}+ T cells transduced with the 6F9 TCR but not the R12C9 TCR recognized the MAGE-A3.sup.+ /HLA-DP*0401.sup.+ melanoma cell line 1359 mel-CIITA but failed to recognize the MAGE-A3.sup.+ /HLA-DP*0401.sup.- melanoma cell line 624 mel-CIITA. CD4^{sup.}+ T cells transduced with the R12C9 TCR failed to recognize either of the tested melanoma cell lines. Cells transduced with the MART-1 reactive TCR DMF5 failed to recognize the transfected 293-CIITA cells or MAGE-A3:sub.243-258 pulsed target cells, but recognized the HLA-A*0201⁺ and MART-1⁺ cell line 624 mel-CIITA. The experiments were repeated using PBL from a third human donor and similar results were obtained. Because the 6F9 TCR was obtained from a Treg clone, which are involved in the suppression of immune activity, the reactivity of the 6F9 TCR was surprising and unexpected. These results indicated that while CD4^{sup.}+ T cells transduced with 6F9 or R12C9 recognized

peptide pulsed target cells, only cells transduced with the 6F9 TCR recognized transfected target cells as well as MAGE-A3.sup.+ and HLA-DP*04.sup.+ tumor cells.

EXAMPLE 3

(117) This example demonstrates that 6F9-transduced PBLs show high reactivity to MAGE-A3 full-length protein processed and presented by HLA-DP4+ B cells.

(118) PBL from two human donors was untransduced or transduced with a nucleotide sequence encoding the 6F9 TCR. The cells were co-cultured with HLA-DP4+ B cells that had processed and presented full-length MAGE-A3 protein (SEQ ID NO: 1). The results are shown in Table 3 and FIG. 10A. As shown in Table 3 and FIG. 10A, the 6F9-transduced PBLs were highly reactive to MAGE-A3 full-length protein processed and presented by HLA-DP4+ B cells.

(119) TABLE-US-00004 TABLE 3 Donor 1 IFN-γ (pg/ml) MAGE-A3 Untransduced 6F9 TCR DP4 (μg/ml) (UT) transduced B cells + MAGE-A3 + 10 360 23640 full length + 1 420 12440 + 0.1 358 2360 + 0.01 362 580 + 0.001 343 427 + 0.0001 349 387 + 0 343 405 B cells + NY-ESO-1 + 10 313 363 full length Donor 2 IFN-γ (pg/ml) MAGE-A3 Untransduced 6F9 TCR DP4 (μg/ml) (UT) transduced B cells + MAGE-A3 + 10 2080 63100 full length + 1 1810 21270 + 0.1 1382 3590 + 0.01 1519 685 + 0.001 1297 470 + 0.0001 1568 542 + 0 1351 404 B cells + NY-ESO-1 + 10 1549 530 full length

EXAMPLE 4

(120) This example demonstrates that 6F9 TCR-transduced PBLs are reactive to tumor lines with endogenous class II presentation of MAGE-A3 protein.

(121) PBL from two human donors were untransduced or transduced with a nucleotide sequence encoding the 6F9 TCR or F5 TCR. The cells were cultured alone (T cell only) or co-cultured with 624-CIITA cells, 526-CIITA cells, or H1299-CIITA cells (tumor cell lines transfected with CIITA). The results are shown in Table 4. As shown in Table 4, 6F9 TCR-transduced PBLs were reactive to tumor lines with endogenous class II presentation of MAGE-A3 protein.

(122) TABLE-US-00005 TABLE 4 IFN-gamma (pg/ml) Donor 1 Un- F5 6F9 DP4 MAGE A3 transduced transduced transduced 624-CIITA - + 223 1483 238 526-CIITA + (DP4 0401) + 636 2360 1314 H1299- + (DP4 0401) + 284 243 4330 CIITA T-cell only 131 45 112 IFN-gamma (pg/ml) Donor 2 Un- F5 6F9 DP4 MAGE A3 transduced transduced transduced 624-CIITA - + 819 1435 153 526-CIITA + + 117 2530 1339 H1299- + + 147 172 3630 CIITA T-cell only 65 27 88

EXAMPLE 5

(123) This example demonstrates that the 6F9 TCR is MAGE-A3 specific.

(124) PBL from a human donor were CD4+ enriched and the number of cells was rapidly expanded on day 27. Cells were untransduced or transduced with F5 TCR or 6F9 TCR and co-cultured with 526-CIITA cells or H1299-CIITA cells alone or with anti-MAGE-A3 siRNA or anti-MART-1 siRNA. IFN-gamma secretion was measured. The results are shown in FIGS. 2A and 2B.

(125) As shown in FIGS. 2A and 2B, the anti-MAGE-A3 siRNA reduced the reactivity of the 6F9-TCR transduced cells. Accordingly, the siRNA knockdown assay confirmed that the 6F9 TCR is MAGE-A3 specific.

EXAMPLE 6

(126) This example demonstrates that 6F9 TCR recognizes MAGE-A3 in an HLA-DP restricted manner.

(127) 624, 526, 1359, H1299, 1300, 1764, 3071, 397, 2630, and 2984 tumor cell lines were transduced with CIITA (624-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, 1300-CIITA, 1764-CIITA, 3071-CIITA, 397-CIITA, 2630-CIITA, and 2984-CIITA) and HLA-DP expression was measured by flow cytometry. DP4 and MAGE-A3 expression is shown in Table 5A.

(128) TABLE-US-00006 TABLE 5A Transduced Tumor Cell Line DP4 MAGE-A3 624-CIITA — + 1300-CIITA — + 3071-CIITA 0402 + Whittington-CIITA 0401 - 526-CIITA 0401 + 1359-CIITA 0401 + H1299-CIITA 0401 + 397-CIITA 0401 + 2630-CIITA 0401 + 2984-CIITA 0401 +

(129) 6F9-transduced PBL were cultured alone (T cells only) or co-cultured with 3071 cells, 3071-

CIITA cells, 397 cells, 397-CIITA cells, 2630 cells, 2630-CIITA cells, 2984 cells, and 2984-CIITA cells. IFN-gamma secretion was measured. The results are shown in FIG. 3. As shown in FIG. 3, 6F9-transduced PBL were reactive with CIITA-expressing tumor cell lines.

(130) The 6F9 TCR was further evaluated by determining the reactivity of CD4.sup.+ and CD8.sup.+ T cells separated from two patient PBMCs against a panel of tumor cell lines including 624-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, SK37-CIITA, 1764-CIITA, 3071-CIITA, 397-CIITA, 2630-CIITA, and 2984-CIITA. Five melanoma cell lines that expressed MAGE-A3 and HLA-DP*0401 (2630-CIITA, 397-CIITA, 2984-CIITA, 526-CIITA, and 1359-CIITA), as well as the non-small cell lung carcinoma cell line H1299 NSCLC-CIITA were recognized by transduced CD4.sup.+ and CD8.sup.+ T cells, although CD4.sup.+ T cells secreted higher amounts of cytokine in response to tumor targets than transduced CD8.sup.+ T cells.

(131) H1299-CIITA and 526-CIITA cells were transfected with anti-HLA-DP or anti-HLA-DR siRNA to knock down HLA-DP or HLA-DR expression. 3071-CIITA and 526-CIITA cells were transfected with anti-HLA-DQ siRNA to knock down HLA-DQ expression. HLA-DP, HLA-DR, or HLA-DQ knockout was confirmed by flow cytometry.

(132) PBL from a human donor were enriched for CD4+ and the number of cells was rapidly expanded on day 30. The cells were transduced with 6F9 TCR or untransduced. The cells were cultured alone (T cell only) or co-cultured with untreated H1299-CIITA cells, H1299-CIITA transfected with anti-HLA-DP or anti-HLA-DR siRNA, untreated 526-CIITA cells, or 526-CIITA transfected with anti-HLA-DP or anti-HLA-DR siRNA. IFN-gamma secretion was measured. The results are shown in FIG. 4. As shown in FIG. 4, the anti-HLA-DP siRNA reduced the reactivity of the 6F9-TCR transduced cells.

(133) Further studies employing antibodies confirmed that the 6F9 TCR recognizes MAGE-A3 in an HLA-class II restricted manner. PBL transduced with 6F9 TCR were co-cultured with the cells set forth in Table 5B and blocked with the antibodies set forth in Table 5B. IFN-gamma was measured, and the results are set forth in Table 5B.

(134) TABLE-US-00007 TABLE 5B 6F9 TCR-transduced PBL IFN-gamma co-cultured with:
Blocked with antibody: (pg/ml) 293-CIITA (DP4+) W6/32 (α-HLA class I) >10,000 transfected with MAGE-A3 HB22 (α-HLA class DR) >10,000 gene IVA12 (α-HLA class II) 902 Allen B cells (A2+ DP4+) W6/32 (α-HLA class I) 15038 Incubated with MAGE-A3 HB22 (α-HLA class DR) 16599 protein IVA12 (α-HLA class II) 129 SK37 CIITA (A2+ DP4+ W6/32 (α-HLA class I) 1965 MAGE-A3+) HB22 (α-HLA class DR) 6248 IVA12 (α-HLA class II) 674 HI 299 CIITA (A2-DP4+ W6/32 (α-HLA class I) 2684 MAGE-A3+) HB22 (α-HLA class DR) 7888 IVA12 (α-HLA class II) 0 1764 RCC CIITA (A2-DP4+ W6/32 (α-HLA class I) 0 MAGE-A3-) HB22 (α-HLA class DR) 0 IVA12 (α-HLA class II) 0

(135) As shown in Table 5B, the antibody blocking studies showed that the 6F9 TCR recognizes MAGE-A3 in an HLA Class II-restricted manner, but not in an HLA-DR-restricted manner or in an HLA Class I-restricted manner.

EXAMPLE 7

(136) This example demonstrates that an alanine substitution at position 116 or 117 of the alpha chain of the 6F9 TCR increases the reactivity of the 6F9 TCR.

(137) Eight different substituted TCRs, each having one alanine substitution at a different location in the CDR3 region of the 6F9 TCR, were prepared as set forth in Table 6.

(138) TABLE-US-00008 TABLE 6 Name Description SEQ ID NO: a1 Alanine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 116 of alpha chain (S116A) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ala, Xaa at 117 is Ser, Xaa at 118 is Gly, and Xaa at 119 is Thr a2 Alanine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 117 of alpha chain (S117A) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ser, Xaa at 117 is Ala, Xaa at 118 is Gly, and Xaa at 119 is Thr a3 Alanine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 118 of alpha chain (G118A) SEQ ID NO: 33 (substituted alpha

chain), wherein Xaa at 116 is Ser, Xaa at 117 is Ser, Xaa at 118 is Ala, and Xaa at 119 is Thr a4 Alanine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 119 of alpha chain (T119A) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ser, Xaa at 117 is Ser, Xaa at 118 is Gly, and Xaa at 119 is Ala b1 Alanine substitution at position SEQ ID NO: 11 (wild-type (wt) alpha chain) and 115 of beta chain (R115A) SEQ ID NO: 34, wherein Xaa at 115 is Ala, Xaa at 116 is Thr, Xaa at 117 is Gly, and Xaa at 118 is Pro b2 Alanine substitution at position SEQ ID NO: 11 (wild-type (wt) alpha chain) and 116 of beta chain (T116A) SEQ ID NO: 34, wherein Xaa at 115 is Arg, Xaa at 116 is Ala, Xaa at 117 is Gly, and Xaa at 118 is Pro b3 Alanine substitution at position SEQ ID NO: 11 (wild-type (wt) alpha chain) and 117 of beta chain (G117A) SEQ ID NO: 34, wherein Xaa at 115 is Arg, Xaa at 116 is Thr, Xaa at 117 is Ala, and Xaa at 118 is Pro b4 Alanine substitution at position SEQ ID NO: 11 (wild-type (wt) alpha chain) and 118 of beta chain (P118A) SEQ ID NO: 34, wherein Xaa at 115 is Arg, Xaa at 116 is Thr, Xaa at 117 is Gly, and Xaa at 118 is Ala

(139) PBL from a human donor were untransduced or transduced with wild-type (wt) 6F9 TCR or one of each of the eight substituted TCRs in Table 6. The cells were cultured alone (T cell only) or co-cultured with 624-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, or 1764-CIITA. IFN-gamma secretion was measured. The results are set forth in FIG. 5. As shown in FIG. 5, the a1 and a2 substituted TCRs demonstrated increased reactivity as compared to wt 6F9 TCR.

(140) A separate experiment with transduced, CD4+ enriched PBL also confirmed the superior reactivity of the a1 and a2 substituted TCRs (FIG. 6). As shown in FIG. 6, the a1 and a2 substituted TCRs showed an approximately 2-fold increase in anti-tumor activity as compared to wt 6F9 TCR. The a1 and a2 substituted TCRs also showed better tetramer (SEQ ID NO: 2) binding as compared to the wt 6F9 TCR, as measured by flow cytometry.

EXAMPLE 8

(141) This example demonstrates the reactivity of substituted 6F9 TCRs.

(142) Eight different substituted TCRs, each having one amino acid substitution at a different location in the CDR3 region of the alpha chain of the 6F9 TCR, were prepared as set forth in Table 7.

(143) TABLE-US-00009 TABLE 7 Name Description SEQ ID NO: a1-1 Leucine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 116 of alpha chain (S116L) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Leu, Xaa at 117 is Ser, Xaa at 118 is Gly, and Xaa at 119 is Thr a1-2 Isoleucine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 116 of alpha chain (S116I) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ile, Xaa at 117 is Ser, Xaa at 118 is Gly, and Xaa at 119 is Thr a1-3 Valine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 116 of alpha chain (S116V) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Val, Xaa at 117 is Ser, Xaa at 118 is Gly, and Xaa at 119 is Thr a1-4 Methionine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 116 of alpha chain (S116M) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Met, Xaa at 117 is Ser, Xaa at 118 is Gly, and Xaa at 119 is Thr a2-1 Leucine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 117 of beta chain (S117L) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ser, Xaa at 117 is Leu, Xaa at 118 is Gly, and Xaa at 119 is Thr a2-2 Isoleucine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 117 of beta chain (S117I) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ser, Xaa at 117 is Ile, Xaa at 118 is Gly, and Xaa at 119 is Thr a2-3 Valine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 117 of beta chain (S117V) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ser, Xaa at 117 is Val, Xaa at 118 is Gly, and Xaa at 119 is Thr a2-4 Methionine substitution at position SEQ ID NO: 12 (wild-type (wt) beta chain) 117 of beta chain (S117M) SEQ ID NO: 33 (substituted alpha chain), wherein Xaa at 116 is Ser, Xaa at 117 is Met, Xaa at 118 is Gly, and Xaa at 119 is Thr

(144) PBL from a human donor were untransduced or transduced with wild-type (wt) 6F9 TCR or

one of each of the eight substituted TCRs. The cells were cultured alone (T cell only) or co-cultured with 624-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, or 1764-CIITA. IFN-gamma secretion was measured. The results are set forth in FIG. 7. As shown in FIG. 7, the a1, a2, and a1-3 substituted TCRs demonstrated reactivity against CIITA-tumor cell lines.

EXAMPLE 9

(145) This example demonstrates that substitution of the native constant region of the 6F9 TCR with a murine constant region increases the reactivity of the 6F9 TCR.

(146) A TCR was prepared comprising the variable regions of the α and β chains of the wt 6F9 TCR and a murine constant region (6F9mC TCR) (SEQ ID NOs: 27 and 28).

(147) The 6F9mC TCR demonstrated better MAGE-A3 tetramer and VP staining as compared to wt 6F9 TCR, as measured by flow cytometry. Without being bound to a particular theory, it is believed that the 6F9mC TCR provides improved pairing of the TCR α and β chains.

(148) PBL from a human donor were untransduced or transduced with wt 6F9 TCR or 6F9mC TCR and cultured alone (T-cell only) or co-cultured with 624-CIITA, 1300-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, 397-CIITA, 2630-CIITA, 2984-CIITA, 3071-CIITA, or 1764-CIITA cells. IFN-gamma secretion was measured. The results are shown in FIG. 8. As shown in FIG. 8, the 6F9mC-transduced cells showed a 2-5 fold increase in anti-tumor activity as compared to wt 6F9 TCR-transduced cells.

(149) Untransduced cells, 6F9 TCR-transduced cells, or 6F9mC TCR-transduced cells were enriched for CD8 or CD4 and cultured alone (T-cell only) or co-cultured with 624-CIITA, SK37-CIITA, 526-CIITA, 1359-CIITA, H1299-CIITA, 397-CIITA, 2630-CIITA, 2984-CIITA, 3071-CIITA, or 1764-CIITA cells. Interferon-gamma secretion was measured. The results are shown in FIGS. 9A and 9B. As shown in FIGS. 9A and 9B, the CD8 and CD4 enriched 6F9mC-transduced cells maintained higher anti-tumor activity as compared to 6F9 TCR transduced cells for several cell lines, indicating high affinity of the 6F9mC TCR independent of co-receptors. The experiments were repeated using PBL from a second human donor and similar results were obtained.

Comparisons of responses of CD4⁺ T cells transduced with the wild-type (Wt) 6F9 TCR with those of the cells transduced with the 6F9mc TCR indicated that the murine constant regions resulted in between two and five-fold enhancement in the response of transduced T cells against the seven MAGE-A3.sup.+ and HLA-DP*0401.sup.+ targets that were evaluated. In addition, the response of CD8.sup.+ T cells transduced with the 6F9mc were enhanced by between two and ten-fold above those seen in cells transduced with the wt 6F9 TCR. The responses of CD8.sup.+ T cells transduced with the 6F9mc were generally lower than CD4.sup.+ T cells transduced with this TCR, although comparable cytokine responses were observed in responses to some tumor targets.

EXAMPLE 10

(150) This example demonstrates that upon tumor stimulation, 6F9mC TCR-transduced cells produce high levels of IFN-gamma and TNF-alpha and show a highly activated phenotype (as measured by increased 4-1BB, CD25, and CD69 expression).

(151) Cells were CD4 or CD8 enriched and transduced with 6F9mC TCR. Transduced cells were co-cultured with tumor lines 624-CIITA, 2630-CIITA, 2984-CIITA, or Whittington-CIITA for 6 hours and then stained for intracellular IFN-gamma, interleukin (IL)-2, or tumor necrosis factor (TNF)- α . The 6F9mC TCR transduced cells showed specific intracellular IFN-gamma production upon tumor stimulation. The 6F9mC TCR transduced cells showed detectable IL-2 production and specific high TNF- α production upon tumor stimulation in the CD4-enriched fraction.

(152) Cells were CD4 enriched and transduced with 6F9mC TCR. Transduced cells were co-cultured with tumor lines 624-CIITA, 2630-CIITA, 2984-CIITA, or Whittington-CIITA overnight and then stained for 4-1BB, CD25, and CD69. After overnight tumor stimulation, the majority of 6F9mC TCR-transduced cells expressed high levels of 4-1BB (indicative of antigen-specific activation), CD25, and CD69.

EXAMPLE 11

(153) This example demonstrates that the 6F9 TCR mediates tumor cell recognition.

(154) PBL were untransduced or transduced with wild-type 6F9 TCR and cultured alone or co-cultured with non-small cell lung cancer (NSCLC) cell line H11299 or melanoma cell line 526 mel, 624 mel, or 1359 mel. MAGE-A3 and DP*04 expression is shown in Table 8.

(155) TABLE-US-00010 TABLE 8 Cell line MAGE-A3 DP*04 H11299 NSCLC + + 526 mel + + 624 mel + - 1359 mel + -

(156) IFN-gamma expression was measured. The results are shown in FIG. 10B. As shown in FIG. 10B, the 6F9 TCR mediates tumor cell recognition.

EXAMPLE 12

(157) This example demonstrates that the 6F9 and 6F9mc TCR possess a high degree of specificity for the MAGE-A3::sub.248-258 peptide.

(158) In order to evaluate the fine specificity of antigen recognition mediated by cells transduced with the 6F9 and 6F9mc TCR, HLA-DP*0401.sup.+ target cells were pulsed with truncations of the MAGE-A3::sub.243-258 peptide or related peptides from MAGE family members. CD4+ T cells isolated from two patients' PBMC (PBMC-1 or PBMC-2) by negative selection were transduced with either the 6F9 TCR, the 6F9mc TCR, or were un-transduced and assayed 10 days following OKT3 stimulation for their response to 293-CIITA cells that were pulsed with 10 mg/ml of the peptides indicated in Table 9.

(159) Analysis of the response to truncated MAGE-A3 peptides from two cultures of transduced CD4.sup.+ PBMC indicated that the 11-mer peptide QHFVQENYLEY (SEQ ID NO: 54) corresponding to amino acids 248-258 of the MAGE-A3 protein represented the minimal peptide that elicited a response comparable to that elicited by the MAGE-A3::sub.243-258 peptide (Table 9). The MAGE-A3::sub.243-258 peptide was predicted using an epitope prediction algorithm to possess a high affinity for HLA-DP*0401, and in addition, recognition of the truncated MAGE-A3 peptides appeared to correlate with T cell recognition (Table 9). Significant recognition was observed for the MAGE-A6::sub.248-258 peptide that contained a single substitution of tyrosine for histidine at position 249, but minimal reactivity was observed against additional members of the MAGE family of gene products that possessed between two and five differences from the MAGE-A3::sub.248-258 peptide. A BLAST search of the NCBI database revealed that the most closely related peptide was derived from the protein necdin. This peptide, which possessed five differences from the MAGE-A3::sub.248-258 peptide, was also not recognized by T cells transduced with the 6F9 or 6F9mc TCR. These findings indicate that the 6F9 TCR possesses a high degree of specificity for the MAGE-A3::sub.248-258 peptide, and suggest that T cells transduced with this TCR may possess little or no cross-reactivity with peptides derived from additional human proteins.

(160) TABLE-US-00011 TABLE 9 PBMC-1 PBMC-2 transduced with: transduced with:
Predicted SEQ Amino None 6F9 affinity Gene (position) ID NO: Acid Sequence 6F9 6F9mc
IFN-g (pg/ml) 6F9mc None (nm) MAGE-A3: 243-258 2 KKLLTQHFVQENYLEY 10,220
15,210 33 10,350 17,520 45 3 MAGE-A3: 243-256 47 KKLLTQHFVQENYL 1,018
1,815 72 1,670 2,490 78 323 MAGE-A3: 243-255 48 KKLLTQHFVQENY
76 137 29 111 117 71 378 MAGE-A3: 243-254 49
KKLLTQHFVQEN 28 0 67 30 39 78 466 MAGE-
A3: 243-253 50 KKLLTQHFVQE 0 40 38 30 45 90
2444 MAGE-A3: 245-258 51 LLTQHFVQENYLEY 9,290 14,970 84 8,920 17,820 74
3 MAGE-A3: 246-258 52 LTQHFVQENYLEY 7,140 12,700 56 9,200 16,170 76
3 MAGE-A3: 247-258 53 TQHFVQENYLEY 6,710 10,600 30 6,810 13,280 41
3 MAGE-A3: 248-258 54 QHFVQENYLEY 6,220 9,000 52 7,400 8,700 56
4 MAGE-A3: 249-258 55 HFVQENYLEY 669 1,643 57 922 2,034
66 5 MAGE-A6: 248-258 56 QYFVQENYLEY 6,440 11,800 54 13,200 8,370
127 3 MAGE-A2/A12: 248-258 57 QDLVQENYLEY 33 66 49

37 56 65 59 MAGE-A4/A9: 249-259 58 QDWVQENYLEY
0 23 32 22 26 62 92 MAGE-A8: 251-261 59
QEWVQENYLEY 43 58 79 39 41 55 87 MAGE-
A1/B4: 241-251 60 QDLVQEKYLEY 129 126 55 108 84 53
16 MAGE-B2: 250-260 61 KDLVQEKYLEY 0 0 43 7
20 38 16 MAGE-B10: 250-260 62 KDLVKENYLEY 22 18
69 28 34 66 105 MAGE-B16: 252-262 63 KDFVKEKYLEY
0 27 16 11 28 42 3 MAGE-C1: 113-123 64
KVWVQEHYLEY 9 0 30 25 27 30 35
MAGE-D4: 300-315 65 RKLITDDFVKQKYLEY 193 234 81 194
268 80 6 MAGE-D2: 413-428 66 KKLITDEFVKQKYLDY 82 43
56 226 223 71 8 MAGE-L2: 582-597 67 KKLITEVFVRQKYLEY
45 56 58 78 107 114 6 MAGE-G1: 220-235 68
KKLITEDFVRQRYLEY 0 29 68 25 33 62 3
Necdin: 237-247 69 EEVQMNYLKY 0 22 59 21 32
83 13 No peptide 0 5 58 15 25 59

(161) All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(162) The use of the terms “a” and “an” and “the” and “at least one” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The use of the term “at least one” followed by a list of one or more items (for example, “at least one of A and B”) is to be construed to mean one item selected from the listed items (A or B) or any combination of two or more of the listed items (A and B), unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

(163) Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. A method of producing an engineered population of human cells, the method comprising: introducing a recombinant expression vector to an isolated population of human cells, wherein the

recombinant expression vector comprises a nucleotide sequence encoding a T-cell receptor (TCR) having antigenic specificity for MAGE-A3.sub.243-258 and MAGE-A6, wherein the TCR comprises the amino acid sequences of: (a) SEQ ID NOs: 3, 4, 6, 7, (i) SEQ ID NO: 29, wherein: Xaa4 is Ser, Ala, Leu, Ile, Val, or Met; Xaa5 is Ser, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Thr, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 30, wherein: Xaa4 is Arg, Ala, Leu, Ile, Val, or Met; Xaa5 is Thr, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Pro, Ala, Leu, Ile, Val, or Met; or (b) SEQ ID NOs: 3-8.

2. The method according to claim 1, wherein the TCR comprises the amino acid sequences of SEQ ID NOs: 3-8.

3. The method according to claim 1, wherein the TCR comprises a murine constant region.

4. The method according to claim 3, wherein the TCR comprises the murine constant region amino acid sequence(s) of SEQ ID NO: 25 and/or SEQ ID NO: 26.

5. The method according to claim 1, wherein the TCR comprises: a first amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 31, wherein: Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 9; and a second amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 32, wherein: Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 10.

6. The method according to claim 5, wherein the TCR comprises the amino acid sequences of SEQ ID NO: 9 and SEQ ID NO: 10.

7. The method according to claim 1, wherein the TCR comprises: a first amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 33, wherein: Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; (ii) SEQ ID NO: 11; and (iii) SEQ ID NO: 27; and a second amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 34, wherein: Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met; (ii) SEQ ID NO: 12; and (iii) SEQ ID NO: 28.

8. A method of producing an engineered population of human cells, the method comprising: introducing a recombinant expression vector to an isolated population of human cells, wherein the recombinant expression vector comprises a nucleotide sequence encoding a polypeptide comprising the amino acid sequences of: (a) SEQ ID NOs: 3, 4, 6, 7, (i) SEQ ID NO: 29, wherein: Xaa4 is Ser, Ala, Leu, Ile, Val, or Met; Xaa5 is Ser, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Thr, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 30, wherein: Xaa4 is Arg, Ala, Leu, Ile, Val, or Met; Xaa5 is Thr, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Pro, Ala, Leu, Ile, Val, or Met; or (b) SEQ ID NOs: 3-8.

9. The method of claim 8, wherein the polypeptide comprises the amino acid sequences of SEQ ID NOs 3-8.

10. The method of claim 8, wherein the polypeptide comprises: a first amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 31, wherein: Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 9; and a second amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 32, wherein: Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 10.

11. The method of claim 10, wherein the polypeptide comprises the amino acid sequences of SEQ ID NO: 9 and SEQ ID NO: 10.

12. The method of claim 8, wherein the polypeptide comprises: a first amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 33, wherein: Xaa116 is Ser, Ala, Leu, Ile,

Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; (ii) SEQ ID NO: 11; and (iii) SEQ ID NO: 27; and a second amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 34, wherein: Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met; (ii) SEQ ID NO: 12; and (iii) SEQ ID NO: 28.

13. A method of producing an engineered population of human cells, the method comprising: introducing a recombinant expression vector to an isolated population of human cells, wherein the recombinant expression vector comprises a nucleotide sequence encoding a protein comprising a first polypeptide chain comprising the amino acid sequences of: (a) SEQ ID NO: 3-5; or (b) SEQ ID NOs: 3, 4, and SEQ ID NO: 29, wherein: Xaa4 is Ser, Ala, Leu, Ile, Val, or Met; Xaa5 is Ser, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Thr, Ala, Leu, Ile, Val, or Met; and a second polypeptide chain comprising the amino acid sequences of: (a) SEQ ID NOs: 6-8; or (b) SEQ ID NOs: 6, 7, and SEQ ID NO: 30, wherein: Xaa4 is Arg, Ala, Leu, Ile, Val, or Met; Xaa5 is Thr, Ala, Leu, Ile, Val, or Met; Xaa6 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa7 is Pro, Ala, Leu, Ile, Val, or Met.

14. The method of claim 13, wherein the first polypeptide chain comprises the amino acid sequences of SEQ ID NO: 3-5 and the second polypeptide chain comprises the amino acid sequences of SEQ ID NO: 6-8.

15. The method according to claim 13, wherein the first polypeptide chain comprises an amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 31, wherein: Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 9; and the second polypeptide chain comprises an amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 32, wherein: Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met; and (ii) SEQ ID NO: 10.

16. The method of claim 15, wherein the first polypeptide chain comprises the amino acid sequence of SEQ ID NO: 9 and the second polypeptide chain comprises the amino acid sequence of SEQ ID NO: 10.

17. The method of claim 13, wherein the first polypeptide chain comprises an amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 33, wherein: Xaa116 is Ser, Ala, Leu, Ile, Val, or Met; Xaa117 is Ser, Ala, Leu, Ile, Val, or Met; Xaa118 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa119 is Thr, Ala, Leu, Ile, Val, or Met; (ii) SEQ ID NO: 11; and (iii) SEQ ID NO: 27; and the second polypeptide chain comprises an amino acid sequence selected from the group consisting of: (i) SEQ ID NO: 34, wherein: Xaa115 is Arg, Ala, Leu, Ile, Val, or Met; Xaa116 is Thr, Ala, Leu, Ile, Val, or Met; Xaa117 is Gly, Ala, Leu, Ile, Val, or Met; and Xaa118 is Pro, Ala, Leu, Ile, Val, or Met; (ii) SEQ ID NO: 12; and (iii) SEQ ID NO: 28.

18. The method of claim 1, wherein the isolated population of human cells is an isolated population of human peripheral blood lymphocytes or an isolated population of human peripheral blood mononuclear cells.

19. The method of claim 1, wherein the isolated population of human cells is an isolated population of human T cells.

20. The method of claim 1, wherein the isolated population of human cells is an isolated population of human CD4⁺ T cells.
